Entanglement Entropy evolution in Shor's algorithm

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Abstract (Italiano)

Negli ultimi anni lo sviluppo del quantum computing e la maggior disponibilità di hardware ha portato sempre più esempi pratici di quantum speedup e quantum advantage. Storicamente, l'algoritmo di Shor è stato uno dei primi algoritmi quantistici in grado di mostrare un'accelerazione rispetto agli algoritmi classici fino ad ora noti. Nonostante queste evidenze sperimentali, le motivazioni profonde dello speedup dato dal quantum computing sono ancora oggi difficili da identificare. La computazione quantistica è intrinsecamente parallela e differisce da quella classica per due principali proprietà: la sovrapposizione degli stati quantistici e la presenza di entanglement.
Mentre le conseguenze del principio di sovrapposizione sono immediate, il ruolo specifico dell'entanglement nel creare quantum advantage viene ancora oggi dibattuto, anche a causa del fatto che sia la definizione formale che il calcolo sperimentale dell'entanglement sfociano spesso in sfide complesse. L'entanglement inoltre introduce instabilità in un sistema aperto, quale un reale computer quantistico: la decoerenza tra qubits porta a una perdita di informazione e può minare alla stabilità di un algoritmo nei NISQ (Noisy Intermediate Scale Quantum) computers.
Questo lavoro è volto al calcolo e analisi dello sviluppo dell'entaglement nell'algoritmo di Shor, specificamente sotto la forma di entanglement entropy. Dopo aver discusso tutti i concetti preliminari e i risultati noti, il nostro lavoro si identifica nella simulazione dell'algoritmo di Shor fino a un massimo di 27 qubits al fine di stimare la distribuzione dell'entanglement su tutte le bipartizioni bilanciate del sistema.
I nostri risultati indicano che l'andamento dell'entanglement entropy è fortemente influenzato dai parametri dell'algoritmo. In particolare, il nostro lavoro rivela un ruolo predominante dell'ordine r nell'identificazione delle dinamiche di entanglement: le distribuzioni per bipartizioni generiche hanno un andamento simile a quello dell'entanglement tra registri nel caso in cui r sia una potenza di due. Qualora invece r non sia una potenza di due, la distribuzione dell'entanglement diviene in media sempre più multipartita al crescere di r e il suo comportamento diviene anche funzione del numero N di cui si vuole trovare la fattorizzazione.
Concentrandoci invece sull'entanglement medio per qubit al termine della modular

exponentiation, troviamo che l'entanglement nel caso di bipartizione tra registri e' in media maggiore rispetto al caso per bipartizione generica, tuttavia la crescita dell'entanglement tra registri è più lenta di quella del caso generico.
I risultati del nostro lavoro indicano che l'entanglement dopo la modular exponentiation consiste di due parti: l'entanglement tra registri, funzione del solo ordine r, e l'entanglement all'interno del target register, il quale dipende da entrambi i parametri r e N e cresce in media con N.

Abstract (English)

In recent years, the advances of quantum computing and the increased availability of quantum hardware allowed us to discover several instances of quantum speedup and quantum advantage. Historically, Shor's algorithm was one of the first quantum algorithms to suggest a speedup with respect to the classical case.
Despite the presence of more and more experimental instances, the deep reasons behind the quantum speedup are still difficult to grasp and define. Quantum computing is inherently parallel and it differs from classical computation in two main aspects: state superposition and existence of entanglement.
The specific role of entanglement in quantum computation is still up to debate, since both the theoretical definition and the experimental evaluation of entanglement often results in a rather complex task. Furthermore, the entanglement introduces instability in the computation: decoherence introduces errors in the real computation and can affect the algorithm results in NISQ (Noisy Intermediate Scale Quantum) computers.
This work aims to compute and analyze the entanglement in Shor's algorithm, specifically through the means of entanglement entropy. After discussing all the preliminary concepts and the known results, our work is focused on simulating Shor's algorithm for up to 27 qubits in order to estimate the distribution of entanglement entropy across all balanced bipartitions of the system.
Our results show that the entanglement entropy behaviour is strongly affected by the algorithm inputs, specifically we discover a strong role of the order r: distributions for generic balanced bipartitions follow a trend similar to the registers bipartition when r is a power of two. When this is not the case, the entanglement distribution get on average more and more multipartite with r and the behaviour of the distributions is affected even by the input number N which we want to factorize.
We look at the averaged entanglement entropy for qubit at the end of modular exponentiation and we find the entanglement for the balanced bipartition to be, on average, larger than the registers bipartition. Despite this, the growth of the balance case is steeper than the register case. Our data shows that entanglement after the

modular exponentiation consists of two distinct components: entanglement between registers, which is a function of r only, and entanglement within the target register, which depends on both r and N and grows with N, on average.

1. Introduction to Quantum Computation

The birth of quantum computing can be traced back to 1980, when Yuri Manin introduce the idea of using quantum particles in order to advance our computing power [1]. In 1982, Feynman developed the theory and popularized the chance of exploiting quantum computers to simulate quantum systems [2].
The field of Quantum Computation and information has grown consistently in recent years, developing both its theoretical aspect, experimental setups and practical applications.
In this introduction we'll give the basic tools to understand and formalize the basis of circuit based quantum computing, both as closed and open quantum systems.
We will then briefly summarize the current applications of quantum computing.

1.1 Circuit computation and state vector formalism

1.1.1 Single Qubit state

A quantum computer is based on quantum information units, the quantum bits or **qubits**. A classical bit is described as a stochastic variable with values in {0,1}, a qubit is a system whose state belongs to an Hilbert space. A single qubit state can be represented as a unitary complex superposition of two states, $|0\rangle$ and $|1\rangle$. This set of quantum states form a one qubit basis, called **computational basis**.
The basis states are represented by column vectors

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \qquad \alpha, \beta \in \mathbb{C}\; s.t.\; |\alpha|^2 + |\beta|^2 = 1,\; |\psi\rangle \in \mathbb{C}^2 \tag{1.1}$$

The state can be represented by a complex column vector

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

According to the laws of quantum mechanics, the only outcomes from a measurement of $|\psi\rangle$ can only be $|0\rangle$ or $|1\rangle$ with probability $|\alpha|^2$ and $|\beta|^2$ respectively.

1.1.2 Single qubit gates

In order to modify the qubit state and define quantum algorithms, we have to introduce the notion of quantum gate.
In classical computing, a gate is an operator which performs binary operations. It is defined by a truth table and it can be expressed by a boolean function.
In quantum computing, a quantum gate U on $|\psi\rangle \in \mathbb{C}^{\mathbb{N}}$ is an operator which brings $|\psi\rangle \xrightarrow{U} |\psi'\rangle \in \mathbb{C}^{\mathbb{N}}$ and it can be represented as a N x N matrix with complex entries.
In order for a quantum gate to be well defined and physically meaningful, three conditions have to be met:

- Linearity: $U(a_1\,|\psi\rangle) + a_2\,|\psi'\rangle = a_1\,U\,|\psi\rangle + a_2\,U\,|\psi'\rangle$.
- The action of the gate must be defined in terms of a basis.
- U must be a unitary operator: $UU^\dagger = U^\dagger U = \mathbb{1}$. Unitarity assures preservation of states norm and therefore the probability normalization. Additionally, the gate must be unitary in order for the computation to be reversible.

An example of single qubit gate is the Hadamard gate H, s.t.

$$|0\rangle \xrightarrow{H} \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$

$$|1\rangle \xrightarrow{H} \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$

and has a matrix representation in the computational basis

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \tag{1.2}$$

The Hadamard gate brings every basis state to a uniform superposition state, such that the probability of measuring a basis state is the same for all basis elements. Another fundamental gate is the NOT gate X, defined by its action:

$$|0\rangle \xrightarrow{X} |1\rangle$$

$$|1\rangle \xrightarrow{X} |0\rangle$$

which can be represented in the computational basis by the first Pauli matrix:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{1.3}$$

It's important to underline that gates are not the only viable way of modifying a quantum state. Gates indeed perform a physical in-time-reversible modification of a quantum state, for example the evolution in time of photon polarization. The unitarity of gates represents a perturbation of the state which evolves in time.
Measurement can be used as a different tool in order to modify a quantum state. It is an instantaneous, irreversible and non-unitary transformation. It can be used to build a 'one way quantum computer', in which the entire quantum computation consists of a sequence of one-qubit measurements on an initial state. Thus it uses measurements as the central computation tool. [3]
The graphical representation of a qubit evolution in a quantum computation instance is called a circuit. It consists of input, wires, gates and measurements.

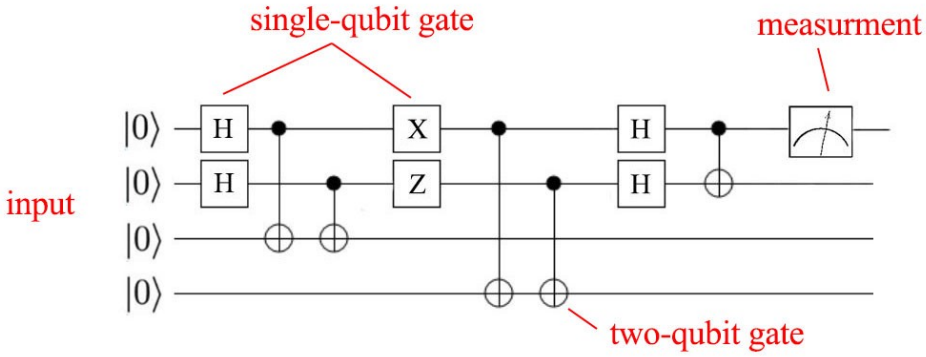


Figure 1.1: Circuit graphical notation

Each line represents a single qubit evolution, starting from an input state, modified by gates (boxes) and measurements.

1.1.3 Multiple qubits state

Given two systems S_1 and S_2, respectively in states $|\psi\rangle_1 \in \mathbb{C}^2$ and $|\psi\rangle_2 \in \mathbb{C}^2$, the overall state $|\psi\rangle \in \mathbb{C}^4$ of the composite system $S = S_1 + S_2$ can be obtained using the tensor product:

$$|\psi\rangle = |\psi\rangle_1 \otimes |\psi\rangle_2$$

As an example, if S_1 is in state $|0\rangle$ and S_2 in state $|1\rangle$, the overall state of S can be expressed as $|0\rangle \otimes |1\rangle = |01\rangle$.
Let $\mathbb{B}(\mathrm{n})$ be the set of all binary strings of length n, $\{|i\rangle\,, i \in \mathbb{B}(\mathrm{n})\}$ is called the computational basis for the n qubits state.
Then $\mathbb{B}(1) = \{0, 1\}$ and $\mathbb{B}(2) = \{00, 01, 10, 11\}$
Using this notation, we can rewrite Equation 1.1 as

$$|\psi\rangle = \sum_{i \in \mathbb{B}(1)} \alpha_i |i\rangle \qquad \alpha_i \in \mathbb{C}\, s.t. \sum_i |\alpha_i|^2 = 1$$

In general, a two qubits system can be represented using the computational basis $\mathbb{B}(2)$ as

$$|\psi\rangle = \sum_{i \in \mathbb{B}(2)} \alpha_i |i\rangle \qquad \alpha_i \in \mathbb{C}\, s.t. \sum_i |\alpha_i|^2 = 1,\ |\psi\rangle \in \mathbb{C}^4$$

Exactly like in the single qubit state, $|\alpha_i|^2$ represents the probability of obtaining the basis element $|i\rangle$ from a measure of state $|\psi\rangle$.
For n qubits, let $N = 2^{\mathrm{n}}$, the state can be expressed as

$$|\psi\rangle = \sum_{i \in \mathbb{B}(\mathrm{n})} \alpha_i |i\rangle \qquad \alpha_i \in \mathbb{C}\, s.t. \sum_i |\alpha_i|^2 = 1,\ |\psi\rangle \in \mathbb{C}^N$$

In a similar way, it's possible to concatenate gates acting on different qubits using the tensor product: gate U_1 acting on $|\psi\rangle_1$ and gate U_2 acting on $|\psi\rangle_2$ give

$$U_1 |\psi_1\rangle \otimes U_2 |\psi_2\rangle = |\phi_1\rangle \otimes |\phi_2\rangle = U |\psi_1\, \psi_2\rangle = |\phi_1 \phi_2\rangle \qquad \text{with } U = U_1 \otimes U_2$$

A peculiar kind of multiple qubit gates are the **controlled gates**, or c-gates.
Given two qubits and a gate U acting on the second qubit, *c-U* is defined as:

$$|00\rangle \xrightarrow{c\text{-}U} |00\rangle$$
$$|01\rangle \xrightarrow{c\text{-}U} |01\rangle$$

$$|10\rangle \xrightarrow{c\text{-}U} |1\rangle \otimes U\,|0\rangle$$
$$|11\rangle \xrightarrow{c\text{-}U} |1\rangle \otimes U\,|1\rangle$$

In this case c-U is acting on the second qubit depending on the first. If U can be written as

$$U = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix}$$

c-U can be represented by

$$c\text{-}U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & u_{00} & u_{01} \\ 0 & 0 & u_{10} & u_{11} \end{bmatrix}$$

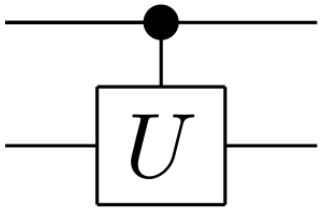


Figure 1.2: Circuit representation of a c-U gate acting on second qubit

As an example, the controlled NOT gate c-NOT is defined as

$$c\text{-}NOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{1.4}$$

The c-NOT gate is essential for quantum computing, since it can be proven that *every multiple qubit gate can be decomposed using the set of all single qubit gates and the c-NOT gate.*
We also define a **quantum register** as an ensemble of qubits in a quantum circuit.

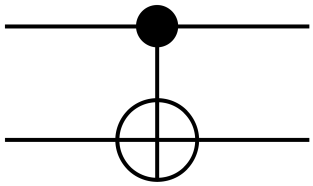


Figure 1.3: Circuit graphical representation of $c\text{-}NOT$ gate

1.2 From vector states to density matrices

1.2.1 Stastical description: density matrices

We discussed gates and states using vector states notation. This is equivalent to assuming we know exactly the state of our qubits and that they have no external interaction. This is not usually the case, as quantum computers and real qubits are usually open quantum systems: qubits interact with the surrounding environment, thus changing their physical state. Furthermore, any physical system will be subject to a nonzero temperature, which will add some statistical noise to the ensemble. This kind of systems cannot be described using vector states because their behaviour has a **statistical component**, while the vector state formalism works for a **deterministic description**.
Consider another phenomenon: let $|\psi\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$ be the state of our system, we perform a measurement of $|\psi\rangle$. Our state will collapse 50% of the time in $|00\rangle$ and the others in $|11\rangle$. If we perform a **non-selective measurement**, i.e. we do not record the collapsed state of our system, we'll have a state with probability one half of being $|00\rangle$ and same of being $|11\rangle$. This situation cannot be described using vector state formalism.
In order to describe the physics of these systems, a new formalism is required.
Let us first assume the state $|\psi\rangle \in \mathbb{C}^n$ of the system is completely known and the system itself is isolated. We can construct a matrix $\rho = |\psi\rangle \langle\psi|$ called **density matrix** where $|\psi\rangle \langle\psi|$ is an outer product written in bra-ket notation. Therefore $\rho \in M_n(\mathbb{C})$, where $M_n(\mathbb{C})$ is the set of all $(n \times n)$ complex matrices.
Using matrix representation for a two qubits state,

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} \qquad \Longrightarrow \rho = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* & \alpha\gamma^* & \alpha\delta^* \\ \beta\alpha^* & |\beta|^2 & \beta\gamma^* & \beta\delta^* \\ \gamma\alpha^* & \gamma\beta^* & |\gamma|^2 & \gamma\delta^* \\ \delta\alpha^* & \delta\beta^* & \delta\gamma^* & |\delta|^2 \end{bmatrix}$$

If a density matrix can be written as a one dimensional projector $P_{\tilde{\psi}} = |\tilde{\psi}\rangle \langle\tilde{\psi}|$ is called a **pure state**. Formally

$$\rho \textbf{ pure state} \iff \exists|\tilde{\psi}\rangle : \rho = |\tilde{\psi}\rangle\langle\tilde{\psi}| \iff \rho^2 = \rho \tag{1.5}$$

A physical system can be described by a pure state if all the uncertainty in the outcomes of ρ comes from the nature of quantum mechanics and therefore we can completely characterize the system by using the vector state $|\tilde{\psi}\rangle$. A ρ which is not pure is called **mixed state**.
An example of mixed state is the non-selective measurement procedure defined above or, in general, an open quantum system.
In open quantum systems, a complete characterization is usually not feasible nor necessary. What we can obtain is a description of the quantum system in terms of a statistical process.
The most general form of a density matrix is

$$\rho = \sum_i \mu_i |\psi_i\rangle \langle\psi_i| \qquad\qquad \sum_i \mu_i = 1,\ \mu_i \geq 0 \tag{1.6}$$

Remark: if $\{|\psi\rangle_i\}_i$ are orthonormal, i.e.

$$\langle\psi_i|\psi_j\rangle = \delta_{ij}$$

then μ_j is the probability of finding the system in state $|\psi\rangle_j$, since

$$Pr(\text{finding } \rho \text{ in state } |\psi_j\rangle) = \langle\rho\rangle_{\psi_j} = \langle\psi_j| \rho |\psi_j\rangle = \sum_i \mu_i |\langle\psi_i|\psi_j\rangle|^2 = \mu_j$$

In the most general case $\langle\psi_i|\psi_j\rangle \neq \delta_{ij}$, then

$$\langle\rho\rangle_{\psi_j} = \sum_i \mu_i |\langle\psi_i|\psi_j\rangle|^2 \neq \mu_j$$

In order for ρ to give an appropriate statistical description of a physical system, it must be a statistical operator, i.e. it must satisfy the following properties:

- Normalization: $\mathrm{Tr}[\rho] = 1$. This condition ensures probabilities are normalized.
- Self-adjoint or Hermitianity: $\rho = \rho^\dagger$. This condition implies that all the eigenvalues (measurable outcomes of a quantum system) are real numbers.
- Semi positive-definiteness: $\lambda_i \geq 0\ \forall i$ with $\{\lambda_i\}_i = \Lambda(\rho)$ is the set of eigenvalues of ρ.

Despite the density matrix formally being an operator, it provides a statistical description of the system, so in quantum open systems the density matrix is usually referred to as state. It encodes the statistics behaviour and evolves according to specific dynamics: Liouville equation for closed system and Linblad for open quantum systems.
From the properties of the trace and the definition of statistical operator, it can be proven that, given an observable A, the mean value of A for a system in state ρ is $< A >_\rho = \mathrm{Tr}[A\rho]$.

1.2.2 Reduced density matrices

Let us now consider a particular framework. Suppose we have a closed quantum system consisting of ten photons described by the density matrix ρ, but due to technical limitations we can only measure a subset S of the system which contains only $u < 10$ photons.
The state of the subsystem S is, in general, affected by the presence of the remaining photons, namely the environment E. This means that the description of S need to take in account the evolution of E.
The problem is similar to the classical case when considering multiple random variables and marginal distributions. The quantum analogous to marginal distributions are called **reduced density matrices**.
A reduced density matrix is a density matrix which effectively describe the behaviour of a subsystem, taking in account its interaction with the surrounding environment. Let ρ be the density matrix of the whole system, we define the reduced density matrix for the subsystem S as the **partial trace** over E of ρ. Formally, let $B(E)$ be a basis for the environment Hilbert Space, then

$$\rho_S = \mathrm{Tr}_E[\rho] = \sum_{e \in B(E)} \langle e| \rho |e\rangle \tag{1.7}$$

where a tensor product with the identity is implied.
The reduced density matrix allows to lower the dimensionality of the problem maintaining measurement results and mean values unchanged. Let A_1 be an observable on S it can be proven that $\forall\, A_1$.

$$\mathrm{Tr}[(A_1 \otimes \mathbb{1})\rho] = \mathrm{Tr}[A_1 \rho_S]$$

The partial trace of a density matrix is still a valid density matrix, because it satisfies normalization, Hermitianity and semi-positive definiteness.

The evolution of a reduced density matrix is non-unitary, since the system is not closed, and therefore the evolution of the state cannot be described by the Liouville equation. The most general master equation known for reduced density matrices is called Linblad equation. [4]

1.3 Current applications of quantum computing

The tools introduced in previous sections are combined in order to develop quantum algorithms.
The most known examples are **Grover's algorithm** and **Shor's algorithm**.
Grover's algorithm is a quantum algorithm for search in an unsorted database. It provides quadratic speedup with respect to the classic counterparts and it is proved to be optimal. It has a variety of applications: it is widely used for brute force optimization, as any discrete optimization problem with no known structure can be translated into a brute search problem. [5]
Shor's algorithm is a phase estimation algorithm which allows to factorize a given integer into prime numbers with an exponential speedup with respect to the classical case. Since a lot of current classical cryptography is based on RSA and prime factorization, this led to a whole new branch of cyber-security, the so called Post Quantum Criptography. [6] [7]
Since quantum computing algorithms can be really useful for general optimization, the use of quantum computers has been established as a resource for machine learning and deep learning. This generated Quantum Machine Learning and other branches like Quantum Optimization, Quantum Natural Language Processing and Quantum Image Processing.
Quantum Reinforcement Learning is also a very peculiar subject, as it consists of two distinct, yet complementary, parts: Quantum Reinforcement Learning can either exploit classical Reinforcement learning in order to design new quantum algorithms or it can consists in the use of Quantum Optimization in Reinforcement Learning training phase. As an instance of the first case, Reinforcement Learning has been applied to generate Quantum error correction codes in Reference[8].
An interesting quantum algorithm for this early stages of quantum development is the Quantum Approximate Optimization Algorithm (QAOA). As the name suggests, it is a quantum optimization algorithm that provides both approximate solutions and a bound for the error.
Since it has been suggested that QAOA may a viable way to prove quantum supremacy [9] on noisy quantum computers, this algorithm has gained further attention.

2. Entanglement and Entanglement measures

In order to understand the motives and implications of this work,in this Chapter we will define and give a physical explanation of entanglement. We will briefly introduce the notions of independence, entropy and mutual information for classic random variables and we will use these concepts to draw a parallel with the quantum case and explain the physical meaning of entanglement.
We will then briefly introduce some measures of entanglement with a focus on entanglement entropy.

2.1 Entanglement definitions for vector states and density matrices

Let $|\psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$ be a vector state representing a physical system S composed of two subsystems S_1 and S_2.
$|\psi\rangle$ can therefore be decomposed using the basis $\{|i\rangle\}_i$ of $\mathcal{H}_1$ and $\{|j\rangle\}_j$ of $\mathcal{H}_2$ as

$$|\psi\rangle = \sum_{i\in\mathcal{H}_1, j\in\mathcal{H}_2} c_{i,j}\,|i\rangle \otimes |j\rangle \tag{2.1}$$

$|\psi\rangle$ is said to be **separable** if $\exists\, |\phi\rangle_1 \in \mathcal{H}_1, |\phi\rangle_2 \in \mathcal{H}_2$ such that

$$|\psi\rangle = |\phi\rangle_1 \otimes |\phi\rangle_2 \tag{2.2}$$

States which are not separable are called **entangled**.
For density matrices, the definition is similar: let ρ be the density matrix representing the state of S, it can be decomposed as

$$\rho = \sum_{i,j} c_{i,j}\,\rho_{1i} \otimes \rho_{2j} \tag{2.3}$$

A density matrix ρ is separable if $\exists c_i, \{\rho_{1i}, \rho_{2i}\}_i$ s.t.

$$\rho = \sum_i c_i \, \rho_{1i} \otimes \rho_{2i} \qquad \sum_i c_i = 1 \tag{2.4}$$

Otherwise, ρ is called an entangled state.

2.2 Meaning of entanglement

In order to explain the physical meaning of entanglement, we first introduce some tools from classical information theory.

2.2.1 Independence and mutual information in classical information theory

Let X be a discrete random variable on a finite dimensional sample space D_X, assume it has a probability density $p_X(x)$, i.e. a function $p_X : D_X \to \mathbb{R}$ such that

$$Pr(X \in A) = \sum_{x \in A} p_X(X = x) \quad \forall A \subseteq D_X$$

$$\sum_{x \in D_X} p_X(X = x) = 1$$

You can then introduce a measure of uncertainty on X called **Shannon entropy**. It's usually denoted by $H(X)$ and it is defined as

$$[0, \log_2(|D_X|)] \ni H(X) = \sum_{y \in D_X} p_X(y) \, \log_2(p_X(y)) \tag{2.5}$$

where $|D_X|$ denotes the cardinality of the sample space D_X. $H(X)$ depends only on the distribution of X and not on the outcomes of X.
The Shannon entropy quantifies the uncertainty on the outcome of X within the distribution $p_X(x)$. Therefore it is zero for deterministic distributions (Kronecker delta or Dirac delta for continuous case) and achieve its maximum value for the uniform distribution. Using Shannon's theorem, the numerical value of H represent the average number of binary variables needed to identify an outcome of X.
Let X, Y be random variables on the sample spaces D_X, D_Y respectively. Let $Z = \{X, Y\}$ be the combined random variable on $D_Z \subseteq D_X \times D_Y$. Let us assume each of them have a probability density $p_X(x)$, $p_Y(y)$ and $p_Z(z)$ respectively.

Then X and Y are **independent** if and only if the combined density can be factorized, i.e.

$$p_Z(x,y) = p_X(x)\, p_Y(y) \quad \forall (x,y) \in D_Z \tag{2.6}$$

and is written as $X \perp Y$.
Independence implies that the probability distribution of the outcomes for X is not affected by any information received on Y.
In physical terms, if we consider X and Y as distinct subsystems of Z, independence means that a measurement on the subsystem X does not give any information on the state of the other subsystem Y.
Using entropy, mutual information $I[X,Y]$ can be defined as

$$I[X,Y] = H[X] + H[Y] - H[X,Y] \geq 0 \tag{2.7}$$

And it can be proven $X \perp Y \iff I[X,Y] = 0$. Mutual information is symmetric and it measures the amount of uncertainty removed from X once the value of Y is known. Furthermore notice that the non-negativity of the mutual information (obtained by using the Jensen's inequality), can be expressed as

$$H(X,Y) \geq H(X) + H(Y) \tag{2.8}$$

The non-negativity of mutual information implies that **the uncertainty of a classical composite system can never be smaller than the uncertainty of its components**.
We also define the Kullback-Leibler divergence or relative entropy D_{KL} between two distributions $p_X(x)$ and $q_X(x)$ as

$$D_{KL}(p||q) = \sum_{x \in D_X} P_X(x) \log_2 (\frac{P_X(x)}{Q_X(x)} \geq 0 \tag{2.9}$$

This does not define a metric since it is not symmetric nor it follows the triangle inequality, but it can be intuitively perceived as a measure of distance between the two distributions P_X and Q_X.
The mutual information can also be rewritten as

$$I(X,Y) = D_{KL}(P_{X,Y}||P_X\, P_Y) \tag{2.10}$$

so the mutual information intuitively provides a measure of distance between the joint distribution and the independent counterpart.

2.2.2 Von Neumann entropy and Quantum Mutual Information

As we defined in Equation 2.2, the overall state of two systems is an entangled state when it cannot be factorized into two independent states. From this formulation it's easy to draw a parallel between quantum entanglement and the independence condition for classical random variables.
As we said, in classical information theory two variables are independent when outcomes of a variable do not give any information about the state of the other. Furthermore, this means that the comprehensive distribution of the two random variables can be factorized in two marginals distribution.
Similarly in quantum mechanics and in quantum information theory two systems are separable when the action of an observable on a subsystem do not affect the state of the other.
This means that two quantum systems S_1 and S_2 are entangled when the complete description of the state of a system requires knowledge on the state of the other.
Since physically each system is described by measurement and observables, two states are entangled when the action of an observable A_1 acting on S_1 cannot be effectively described without considering the state of the subsystem S_2.
In Quantum Information Theory, both entropy and mutual information are redefined as follows.
Let ρ on $\mathcal{H}$ be a density matrix with eigenvalues $\lambda_i \geq 0$, then the **Von Neumann entropy** of ρ, $S(\rho)$, is defined as

$$S(\rho) = -\operatorname{Tr}[\rho \log_2(\rho)] = -\sum_i \lambda_i \log_2(\lambda_i) \qquad \text{with } 0 \log_2(0) = 0 \tag{2.11}$$

Since $S(\rho)$ depends only on the spectrum of ρ, it is basis independent, it is positive (since $\lambda_i \leq 1$) and has an upper bound

$$S(\rho) \leq \log_2(\dim(\mathcal{H})) \tag{2.12}$$

This bound comes directly from the properties of the function $y =$ - x $\log_2(x)$ for $x \in [0, 1]$.
The Von Neumann entropy quantifies the uncertainty within the description of the statistical operator ρ (in bits, since we used base 2 logarithm). A system in a state ρ such that $S(\rho) = 0$, for example, represents a system whose state can be completely characterized. Therefore ρ can be described in a deterministic framework.

As we said, if a quantum system can be described as a deterministic system, its state can be written in the vector state formalism. Therefore

$$S(\rho) = 0 \implies \exists\, |\psi\rangle \text{ s. t. } \rho = |\psi\rangle\langle\psi| \iff \rho \text{ pure}$$

If we focus on pure states, $S(\rho)$ is consistent in highlighting the presence of entanglement, since $S(\rho) = 0 \iff \rho$ pure.

Theorem: $S(\rho) = 0 \iff \rho$ pure state

Proof

- ρ pure $\implies S(\rho) = 0$.

  The matrix $\rho = |\psi\rangle\langle\psi|$ has rank 1 because every row corresponds to $c_i\langle\psi|$ for some coefficient c_i. So the matrix ρ has only one nonzero eigenvalue λ^*. Trace normalization property of density matrices ensures

  $$\sum_{\lambda_i \in \Lambda(\rho)} \lambda_i = 1$$

  Since $\lambda_i = 0\ \forall \lambda_i \neq \lambda^*$, λ^* must be equal to 1.

  Then

  $$S(\rho) = 1\ \log_2(1) + 0\, log(0) + ... + 0\ \log_2(0) = 0$$

- $S(\rho) = 0 \implies \rho$ pure.

  Consider the definition of Von Neumann in terms of eigenvalues in Equation 2.11: each term of the sum is non-negative and it's zero only when an eigenvalue is either 0 or 1.

  Since the sum of non-negative terms is zero, each term must be 0, so $\lambda_i \in \{0, 1\} \forall \lambda_i \in \Lambda(\rho)$. Due to trace normalization, $\Lambda(\rho) = \{1, 0, .., 0\}$. Therefore using as basis the eigenvectors of ρ, ρ is a diagonal matrix with only one nonzero entry, which is equal to 1. Therefore $\rho^2 = \rho$ and ρ is pure as in Equation 1.5.

As a more general statement, given a state ρ_a with entropy S_a, the quantity S_a **represents the minimal number of ancillary qubits we need to entangle with the system a in order to obtain ρ_a as a reduced density matrix of a pure state**.
So given $\rho_a \in M_n(\mathbb{C})$, $S_a = S(\rho_a) = s \in \mathbb{R}$, let $u \in \mathbb{N}$ such that $s \leq u < s+1$, then $\exists\, |\psi\rangle \in \mathbb{C}^{n \cdot u}$ such that $\rho = |\psi\rangle \langle\psi| \in M_{n \cdot u}(\mathbb{C})$ and ρ_a is the reduced density matrix of ρ for the system a.

We now consider a system S that can be divided into two subsystems S_1 and S_2 such that independent measurements can be made on either part. Let $\rho_{1,2}$ be the overall density matrix, ρ_1 the reduced density matrix of S_1 and ρ_2 the complementary density matrix. Using Von Neumann entropy, quantum mutual information can be defined as $I(S_1, S_2)$

$$I(S_1, S_2) = S(\rho_1) + S(\rho_2) - S(\rho_{1,2}) \geq 0 \tag{2.13}$$

where the inequality is due to the subadditivity of Von Neumann entropy, proven in [10]:

$$S(\rho_{1,2}) \leq S(\rho_1) + S(\rho_2) \tag{2.14}$$

This property of the Von Neumann entropy is strictly related to entanglement. In the classical case, the uncertainty of a composite system is always larger than the uncertainty of its components, as in Equation 2.8. In the quantum case though, this constraint is no more present: there are composite states whose entropy is smaller than any of the reduced density matrices.
As an example, consider the two qubits Bell State

$$|\psi_b\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \implies \rho_b = \frac{1}{2}\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

with $S(\rho_b) = 0$. If we consider either reduced density matrix, they are

$$\rho_{red} = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

whose entropy $S(\rho_{red}) = 1$ saturates the bound in Equation 2.12.
We can also link directly the quantum mutual information to the Von Neumann entropy. If we consider a pure state ρ for the system A, $S(\rho) = 0$. Furthermore,

using an important result known as Schmidt decomposition, it can be proved that given A_1, A_2 complementary subsystems of A, the reduced density matrices are equal, i.e. $S(\rho_1) = S(\rho_2)$. This symmetric property holds if and only if the density matrix ρ associated to A is a pure state.
The Von Neumann entropy of a reduced density matrix ρ_{red} is called **entanglement entropy**.
We can conclude that $I(S_1, S_2) = 2\,S(\rho_1)$: given that the overall system is in a pure state, **the entanglement entropy of a subsystem is directly related to the mutual information between the subsystem and its complementary**, therefore it quantifies the correlation between the two subsystems.
We define a new quantity called Quantum Relative entropy between two density matrices ρ and σ as

$$S(\rho||\sigma) = \mathrm{Tr}[\rho\,\log_2(\sigma)] - S(\rho) \geq 0 \tag{2.15}$$

where the inequality above is saturated if and only if $\rho = \sigma$. $S(\rho||\sigma)$ is a measure of distinguishability between the two quantum states ρ and σ and it is the quantum equivalent to the Kullbalck Leibler divergence in Equation 2.9.
Using the definition of quantum relative entropy in Equation 2.15, we can rewrite the quantum mutual information for a generic state as

$$I(S_1, S_2) = S(\rho_{1,2}||\rho_1 \otimes \rho_2)$$

2.3 Entanglement measures for bipartite states

Until now, we have mainly considered entanglement as binary feature: a system can be either entangled or separable. Still, there are various degrees of entanglement and we would like to quantify this physical property.
Various measures of bipartite entanglement have been proposed throughout the years. In this section we will first define the conditions for defining an entanglement measure and we will briefly summarize some important measures.
A bipartite entanglement measure is a function $E(\rho)$ such that

1. $E(\rho) \geq 0 \, \forall \rho$. A normalization factor is usually included such that the maximal entanglement measures for an n qubits systems is $E_{max} = n$.
2. $E(\rho) = 0$ if ρ separable
3. No increase under $LOCC$ (Local Operation Classical Communication)[1]: applying local operations onto ρ and classically communicating do not increase entanglement: let A_{LOCC} be an arbitrary $LOCC$ operation, then $E(\rho) \geq E(A_{LOCC}(\rho))$.
4. Additivity on pure states: $E(\rho^{\otimes n}) = n \, E(\rho)$
5. Asympototical Continuity for pure states [11]:
   Let $\{\rho_n\}_n, \{\sigma_n\}_n$ be sequences of states on a sequence of Hilbert spaces (H_n)
   $$\mathrm{Tr}[|\rho_n - \sigma_n|] \to 0 \implies \frac{E(\rho_n) - E(\sigma_n)}{1 + \log_2[\dim(H_n)]} \to 0$$
6. Subadditivity: $E(\rho \otimes \sigma) \leq E(\rho) + E(\sigma) \quad \forall \rho, \sigma$
7. Convexity: $E(\sum_i p_i \rho_i) \leq \sum_i p_i E(\rho_i)$

Other properties can be added or useful, but not all entanglement measures actually satisfy them. Even the axiomatic definition of entanglement measure can vary between authors, as in Reference [11] and Reference [12].
Even though the meaning and numerical value of entanglement measures can vary, **any entanglement measures that satisfies the properties 1 to 5 is equal to the entanglement entropy on pure states**. The proof is known as "Uniqueness theorem for entanglement measures" and is contained in Reference [11].
We will now list some of the most used entanglement measures.

[1]Local Operations are operation which only acts on a subsystem and can only transform a pure state to another pure state. For a more in depth discussion, see [7]

- Distillable Entanglement $E_D(\rho)$.

  In order to define a measure of entanglement, we first have to fix a scale. We consider as standard unit of entanglement the 2 qubits Bell state $|\psi\rangle^+$

  $$|\psi\rangle^+ = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

  and define $\Phi(K)$ as the maximally entangled density matrix in K dimensions.

  We fix a distance between arbitrary density matrices μ and σ: the trace norm distance $D_1(\mu, \sigma)$

  $$D_1(\mu, \sigma) = \mathrm{Tr}[|\sigma - \mu|] \qquad \text{with } |\rho| = \sqrt{\rho^\dagger \rho}$$

  Consider a two party communication between Alice and Bob and let Alice have n copies of a state ρ, globally $\rho^{\otimes n}$. Since using $LOCC$ it's possible to bring any mixed state to any other mixed state, we can use $LOCC$ transformations Ψ to bring $\rho^{\otimes n} \to \Psi(\rho^{\otimes n}) = \tilde{\rho} \approx \Phi(2^m)$, where the approximation is taken with respect to the distance D_1.

  If for $n \to \infty$ and for fixed rate $r = m/n$, the approximation becomes arbitrarily good, the rate r is called achievable. We can define a measure of entanglement as the greatest achievable rate.

  This measure is called Distillable Entanglement $E_D(\rho)$ and is formally defined as

  $$E_D(\rho) = sup\{r = \frac{m}{n} : \lim_n \inf_{\Psi \in LOCC} \mathrm{Tr}[|\Psi(\rho^{\otimes n}) - \Phi(2^m)|] = 0\}$$

  The operational meaning of this quantity can be explained in the following way: given n copies of a state ρ with $n \gg 1$, $n \cdot E_D(\rho) = m$ represents the highest number of Bell states the state ρ can be transformed into with high fidelity using only $LOCC$ operations.

- Entanglement cost $E_C(\rho)$.

  This measure is conceptually very similar to E_D, but instead of producing Bell states starting from ρ, the entanglement cost $E_C(\rho)$ is defined as the greatest achievable rate at which we can transform m Bell states into n copies of ρ using only $LOCC$, for $n \to \infty$.

Formally,

$$E_C(\rho) = sup\{r = \frac{m}{n} : \lim_{n} \inf_{\Psi} \mathrm{Tr}[|\rho^{\otimes n} - \Psi(\Phi(2^m))|] = 0\}$$

- Entanglement of Formation E_F.

  Let ρ be a mixed state, it can be in general decomposed into different pure states decomposition, called ensembles.

  $$\rho = \sum_i p_{il} \left|\psi_{il}\right\rangle \left\langle\psi_{il}\right|$$

  where i is the index of the i-th element of a specific ensemble $\{p_l, |\psi_l\rangle\}$ and l is the index of the ensemble considered.

  then

  $$E_F(\rho) = \inf_l \{\sum_i p_{il} S(\left|\psi_{il}\right\rangle \left\langle\psi_{il}\right|)\}$$

  where $S(|\psi\rangle \langle\psi|)$ is the entanglement entropy for the state $|\psi\rangle \langle\psi|$ and the bipartition considered.

  The Entanglement of Formation represents the minimum averaged entanglement entropy with respect to all the possible ensembles $\{p_l, |\psi_l\rangle\}$ that can generate ρ (i.e. such that $\rho = \sum_i p_{il} |\psi_{il}\rangle \langle\psi_{il}|$).

- Relative entropies of entanglement $E_R^X(\rho)$.

  Using quantum relative entropy in Equation 2.15 and fixing X, a subset of all the possible states, it is possible to define the Relative entropy of entanglement E_R^X as

  $$E_R^X = \inf_{\sigma \in X} S(\rho||\sigma)$$

  This quantifies a distance between the state ρ and a chosen set of states X. In order for this measure to make sense, the reference set X must be closed under LOCC operations.

  One of the most used reference set is the set of separable states X^*, which is also convex. This makes the underlying optimization problem easier to solve and $E_R^{X^*}$ has a straightforward interpretation: it defines entanglement as the minimal distance of the state ρ from the set of separable states.

- Entanglement entropy $E(\rho) = (S \circ \mathrm{Tr}_E)(\rho)$, which has already been discussed. It is often used in Quantum Field Theory and, as a generalization of Shannon's entropy, it arises as a natural measure of entanglement in Quantum Information Theory.

Since we are gonna consider bipartite entanglement on pure states, each of these measures is equivalent and they are all equal to the entanglement entropy. Entanglement entropy is usually preferred for numerical tasks since it does not involve optimization (see E_R^X, E_F) nor variational definitions (see E_D, E_C), but only requires spectral decomposition of the reduced density matrix, which can be obtained using fast algorithms, optimized for Hermitian matrices.
Another issue in computing most entanglement measures is their **lockability**: most of the entanglement measures can arbitrarily decrease when a single qubit is discarded. Entanglement cost, Entanglement of Formation and Distillable entanglement are lockable, while Relative entropies of entanglement are nonlockable[13].
Lockability prevents us to lower the dimensionality of the problem while retaining a bounded precision: since discarding a single qubits can arbitrarily decrease the value of the entanglement measure, in general we cannot project the ρ into a lower dimensional subspace, because we'd have no bound on the error.
All the entanglement measures listed here are defined for bipartite systems in both pure and mixed states, while their generalization to a multipartite system is often non-trivial: the definition of a consistent entanglement measures for multipartite systems is an active branch of research [14][15].

2.4 Entanglement as a measure of computational complexity

Despite the formal definition of entanglement being complex and not unique, both for mixed and multipartite states, we are still able to use entanglement as a measure of complexity.
We first briefly introduce the Schmidt theorem for a composite state; we suggest Reference [7] for a proof of the theorem.

Schmidt theorem
Let $|\psi\rangle$ be a composite state in $\mathbb{H}_1 \otimes \mathbb{H}_2$, there exists bases $\{|u_i\rangle\}_i$ and $\{|v_i\rangle\}_i$ of $\mathbb{H}_1$ and $\mathbb{H}_2$ respectively, such that

$$|\psi\rangle = \sum_{i=1}^{m} \sqrt{\mu_i}\, |u_i\rangle \otimes |v_i\rangle$$

where $m = \min(\dim(\mathbb{H}_1), \dim(\mathbb{H}_2))$.

The coefficients μ_i in Schmidt theorem are called Schmidt coefficients and from the theorem it follows

- $|\psi\rangle$ is separable $\iff \exists! j : \mu_j \neq 0$, i.e. there's only a nonzero Schmidt coefficient $\mu_j = 1$.
- If more than a coefficient μ_i is nonzero, the state $|\psi\rangle$ is entangled. The number of nonzero Schmidt coefficients is called **Schmidt rank** and we will denote it by χ.
- If all Schmidt coefficients are nonzero and equals, $|\psi\rangle$ is called maximally entangled state.

We can therefore use the Schmidt rank χ as another measure of entanglement, as for example in Reference [16]. There's also a direct relation between entanglement entropy and Schmidt decomposition: the eigenvalues of a reduced density matrix are equal to the square root of the Schmidt coefficients.
In Reference [17], the author proved that entanglement, specifically quantified through the Schmidt rank, can be used in order to assess whether exponential speedup is achievable for a quantum algorithm.

Let us take a system of n qubits, define R as the maximal Schmidt rank over all possible bipartitions. In Reference [17] it has been proven that in order for a quantum algorithm to provide exponential speedup, $R \notin O(poly(n))$ must occur at some point in the computation.
The statement of the theorem is the following: if R is at most polynomial in n throughout the whole quantum computation, the algorithm can be efficiently simulated using a polynomial amount of classical resources (memory and time).
Since for any fixed bipartition holds $S(|\psi\rangle) \leq \log_2(R)$, using this bound between entanglement entropy and maximal Schmidt rank we can conclude that **if a quantum algorithm provides exponential speedup, then at some point of the computation $S(|\psi\rangle) \approx n^b$ for some positive constant b.**
Therefore, a necessary condition for exponential speedup is for entanglement entropy to be polynomial in the number of qubits.

3. Shor's algorithm

Shor's algorithm is one of the most famous quantum algorithms: its aim is to factor a given integer $N \in \mathbb{N}$. A single successful application of the algorithm will return a non-trivial divisor $a \in \mathbb{N}$ s.t. $a \notin \{1, N\}$ and $p \in \mathbb{N}$ where $p = a/N$.

Since a number N has at most a logarithmic amount of prime factors, successfully iterating the algorithm a logarithmic number of times will give the full factorization of N.

The relevance of this algorithm relies on two aspects: firstly, since it provides an exponential speedup with respect to every known classical counterpart, Shor's algorithm was historically one of the first evidence suggesting that quantum computers can solve problem which classical computers cannot. On the other hand, there's no theoretical proof that the factoring problem cannot be solved in polynomial time using a classical computer, even though a solution with non-exponential complexity is yet to be found.

Instead, other quantum algorithms have been found which are proven to provide exponential speedup even when compared to classical probabilistic computation, as in Reference [18].

Furthermore, other algorithms are now being considered as more effective for proving the advantages and alleged supremacy of quantum computing. An example is given by the Quantum Approximate Optimization Algorithm, or QAOA [9], which in recent times it's been in the midst of a debate.

Shor's quantum algorithm is extremely important because its sole existence created a whole new branch of research: post-quantum cryptography. Due to cyber-security and RSA keys being based on classical computers' inability of solving the factoring problem, the discovery of this algorithm raised the urge to find new secure protocols, which could resist an attack even from quantum computers.

We will first describe briefly Shor's algorithm and its components. We will then briefly summarize a possible circuit implementation of the quantum part of the algorithm.

3.1 Shor's algorithm and order-finding

We can describe the algorithm by dividing it into subsequent steps. It is important to notice that all steps can be efficiently performed on a classical computer aside from step 4: every classical order finding algorithm currently known is exponential in resources. The quantum algorithm instead needs polynomial resources, therefore it provides an exponential speedup. On the other hand, it has not been proven whether a non-exponential classical algorithm can exist.
Shor's algorithm is defined as follows.

Shor's algorithm

1. Check if N is even. If it is, return 2.
2. Check if N is a prime power:$\exists p, l \in \mathbb{N}\, s.t.\, p^l = N$ with p prime. If it is, return p.
3. Take a random integer $Y \in \{2.., N-1\}$. If $GCD(Y, N) = a$ is non trivial, i.e. $a > 1$, return a. Otherwise, $\exists R \subset \mathbb{N} : \forall r_i \in R, r_i > 1$, and $Y^{r_i} \bmod N = 1$. The order of Y modulo N is defined as $r(Y, N) = \min_i r_i$.
4. Find the order r of Y modulo N.
5. If r is even, i.e. $r = 2s$, then $Y^{2s} = 1 \bmod N$, which implies $Y^{2s} - 1 = (Y^s + 1)(Y^s - 1) = 0 \bmod N$. Since this is equivalent to $(Y^s + 1)(Y^s - 1) = mn$ for some $m \in \mathbb{N}$, we know either $(Y^s + 1)$ or $(Y^s - 1)$ share a non-trivial common divisor with N. Compute $GCD(Y^s \pm 1, N) = h$ and return it if $h > 1$.
6. If r is odd or $h = 1$ (which may happen when either $Y^s + 1$ or $Y^s - 1$ are equal to a N or a multiple of N), the procedure fails and a new Y is selected.

Repeating the algorithm whenever it fails is still efficient since the probability of success it's greater than $1/2$ for Y, N coprimes, as shown in Reference [6].
The quantum order-finding subroutine in step 3 needs a circuit with two registers, which are called **control** and **target register**. The latter needs to contain all the

natural numbers in {0,..,N-1}, so it consists in $L = \text{ceil}(\log_2(N))$ qubits, where ceil : $x \in \mathbb{R} \to c \in \mathbb{N} : c - 1 < x \leq c$. We will denote by $|\psi_c\rangle |\psi_t\rangle$ an overall product state of the system, where $|\psi_c\rangle$ is the control register's state and $|\psi_t\rangle$ the target register's.

Order-finding subroutine

1. The circuit starts in the initial state $|0\rangle |1\rangle$
2. By applying the Hadamard gate for $2L$ qubits $H^{\otimes 2L}$ on the control register, create a complete superposition of states in the larger register:

$$|0\rangle |1\rangle \xrightarrow{H^{\otimes 2L}} \frac{1}{2^L} \sum_{l=0}^{2^{2L}-1} |l\rangle |1\rangle$$

3. Apply the modular exponentiation operation on the target register

$$|l\rangle |k\rangle \to |l\rangle |k\, Y^l \bmod N\rangle$$

and get

$$\frac{1}{2^L} \sum_{l=0}^{2^{2L}-1} |l\rangle |Y^l \bmod N\rangle$$

4. Apply the Inverse Quantum Fourier Transform (IQFT) on the control register and obtain

$$\frac{1}{2^{2L}} \sum_{l=0}^{2^{2L}-1} \sum_{m=0}^{2^{2L}-1} exp(-2\pi i l m / 2^{2L}) |m\rangle |Y^l \bmod N\rangle$$

5. Measure the control register in the computational basis and convert the binary string obtained to an in integer $c \neq 0$. Using continued fraction representation, i.e. $\frac{c}{2^{2L}} \approx \frac{k}{r}$ where $k \in \{0, .., r-1\}$ can be obtained classically from the value of c. If r and k have a common divisor p, the correct order is not retrieved, instead r/p is found. If this happens, repeat the algorithm with a different (Y, N) couple.

We also use decimal notations for composite qubits state: let m be the number of system qubits, we denote by $|n\rangle$, $n \in \mathbb{N}$ the binary expansion of n with m qubits written in the computational basis: for a 4-qubits system, $|10\rangle = |1010\rangle$ while for a 5 qubits-system, $|2\rangle = |00010\rangle$.
The size of the control register determines the precision of the algorithm. Often, the preferred size is $2L$ qubits, so we decided to follow this convention.
The order-finding subroutine here has been briefly summarized, for a more in-depth discussion, see Reference [7].
The steps we enumerated are roughly the same for all implementations. Despite this, some steps may be implemented using different gates: as we said, a composite gate can be decomposed in the set of one-qubit gates and $c\text{-}NOT$. On the other hand we can arbitrarily chose a one-qubit basis set of gates in order to obtain equivalent circuits.
In this work we will discuss a particular implementation of the circuit for order-finding subroutine, but we will address equivalent circuits when they will be relevant for our study.

3.2 Shor's circuit: computational steps

A graphical representation of the circuit for order-finding is shown in figure 3.1
We will treat the IQFT as a single gate. We are now interested in describing the modular exponentiation operation as a quantum circuit, i.e. as a sequence of quantum gates.
First, we have to check whether the modular exponentiation can be expressed as a unitary gate.
If $f : \{0, .., N-1\} \to \{0, .., N-1\}$ is a bijective function and let $|x\rangle_{x=0}^{N-1}$ be an orthonormal base for the Hilbert space $\mathbb{H}$, we can define the linear map $U_f : \mathbb{H} \to \mathbb{H}$ as

$$U_f |x\rangle = |f(x)\rangle$$

Since f is bijective, U_f is unitary. Let us consider the modular product function

$$f(x) = xY \bmod N$$

Given an integer N and one of its coprimes Y, this function is bijective on the set

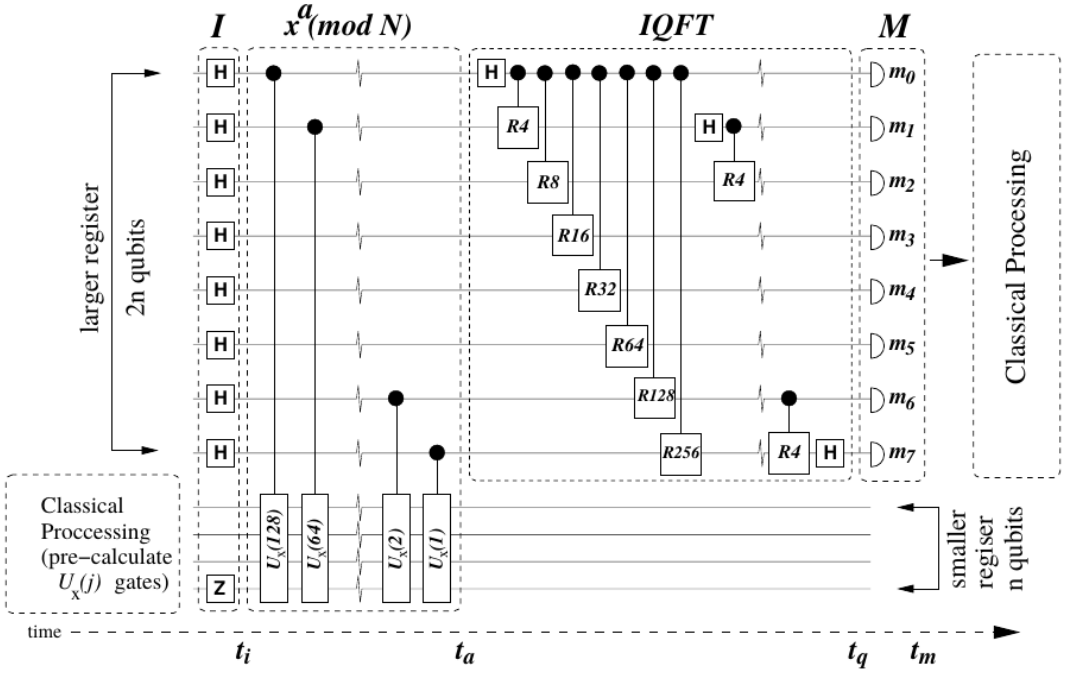


Figure 3.1: Order-finding quantum circuit for a 12 qubits system, i.e. L=4, taken from [19]

$\{0,..,N-1\}$. We can generalize the function on a set $\{0,..,N'-1\}$, $N' > N$ as

$$f(x) = \begin{cases} xY \bmod N & \text{if } 0 \leq x < N \\ x & \text{otherwise} \end{cases}$$

which is bijective on the set $\{0,..,N'-1\}$, therefore the associated linear map U_f is unitary.
We now show how this function can be used to generate the modular exponentiation: we want to obtain $Y^m \bmod N$. We first make a binary expansion of m on t bits

$$m = m_1 2^{t-1} + m_2 2^{t-2} + ... + m_t 2^0 = \sum_{n=1}^{t} m_n 2^{t-n}$$

then

$$Y^m x \;=\; Y^{\sum_{n=1}^{t} m_n 2^{t-n}} x \;=\; \left(\prod_{n=1}^{t} m_n 2^{t-n}\right) x$$
$$=\; Y^{m_1 2^{t-1}}(Y^{m_2 2^{t-2}}...(Y^{m_t 2^0} x)...)$$

We can construct a gate $c\text{-}U_f^{2^k}$, controlled by the control register and applied to the target, such that

$$c\text{-}U_f^{2^k} \left|s\right\rangle \left|x\right\rangle \;=\; \left|s\right\rangle \left|xY^{s2^k} \bmod N\right\rangle$$

We define $U_k = c\text{-}U_f^{2^{k-1}}$ as the gate acting on the target register which has control register qubit $\left|m_{t+1-k}\right\rangle$. We can therefore apply a sequence of gates $U_1, .., U_t$ to the global state $\left|m\right\rangle \left|x\right\rangle$ such that:

$$U_1 \left|m\right\rangle \left|x\right\rangle = U_1 \left|m_1\right\rangle \left|m_2\right\rangle \ldots \left|m_t\right\rangle \left|x\right\rangle = \left|m_1\right\rangle \left|m_2\right\rangle \ldots \left|m_t\right\rangle \left|xY^{m_t 2^0}\right\rangle$$

Continuing the sequence of gates give

$$U_2 U_1 \left|m\right\rangle \left|x\right\rangle = \left|m_1\right\rangle \left|m_2\right\rangle \ldots \left|m_t\right\rangle \left|xY^{m_{t-1}2^1 + m_t 2^0}\right\rangle$$

and the application of all U_k gives the desired modular exponentiation.
We highlight that this implementation of the modular exponentiation operation can be changed: in Figure 3.1 for instance, the order of the U_k gates is swapped. This freedom in implementation does not affect the final result c of the computation, but it changes the intermediate state of the system and, therefore, its entropy.
From now on, we will refer to k as the **computational step** for Shor's algorithm according to this implementation, therefore the state $\left|\psi_k\right\rangle$ of the system after the k-th computational step is

$$\begin{aligned} \left|\psi_k\right\rangle &= U_k \ldots U_2 U_1 \frac{1}{2^L} \sum_{m=0}^{2^{2L}-1} \left|m\right\rangle \left|1\right\rangle \\ &= \frac{1}{2^L} \sum_{m=0}^{2^{2L}-1} \left|m\right\rangle \left|Y^{\sum_{i=0}^{k-1} m_{2L-i} 2^i} \bmod N\right\rangle \end{aligned} \tag{3.1}$$

From this formulation we can also compute analytically the entanglement entropy of every bipartition contained in the target register, as we explain in Appendix A. Although this may lead to further results and computationally feasible simulations, we are not interested in the entanglement of these bipartitions.

4. Entanglement entropy between registers in Shor's algorithm

In this section we will analyze the entanglement entropy between registers in Shor's algorithm. We will first derive a specific formula for the computation of the entropy, then we will analyze and model the obtained results.

4.1 Analytical results

We first consider the first step of the order-finding algorithm

$$|0\rangle |1\rangle \rightarrow |+_{2L}\rangle |1\rangle$$

Both start and end states are product states, so the entanglement entropy between the registers is zero. We then consider the state after the k-th computational step, as in Equation 3.1.
We stress that since the IQFT only acts on the control register, it does not affect the entanglement entropy between registers, so here we will consider only $k \in \{1, .., 2L\}$. If we take $k = 1$

$$|\psi_1\rangle = \frac{1}{\sqrt{2}} |+_{2L-1}\rangle \sum_{m_{2L}=0}^{1} |m_{2L}\rangle |Y^{m_{2L}2^0}\rangle$$

the entanglement may be present only between the last control register qubit and the whole target register. We can rewrite Equation 3.1 as

$$|\psi_k\rangle = \frac{1}{2^{k/2}} |+_{2L-k}\rangle \sum_{m_{2L-k+1}=0}^{1} \cdots \sum_{m_{2L}=0}^{1} |m_{2L-k+1} \dots m_{2L}\rangle |Y^{\sum_{n=0}^{k-1} m_{2L-n}2^n} \bmod N\rangle \tag{4.1}$$

Let $\rho^+ = |+\rangle \langle +|$,then the density matrix at step k is

$$\rho_k = \frac{1}{2^k}(\rho^+)^{\otimes 2L-k} \otimes \sum_{i,j=0}^{2^k-1} |i\rangle \langle j| \otimes |Y^i \bmod N\rangle \langle Y^j \bmod N|$$

We then trace away the control register in order to obtain the reduced density matrix

$$\rho_k^{red} = \frac{1}{2^k} \operatorname{Tr}_c \left[(\rho^+)^{\otimes 2L-k} \otimes \sum_{i,j=0}^{2^k-1} (|i\rangle \langle j|) \otimes |Y^i \bmod N\rangle \langle Y^j \bmod N| \right]$$

The trace do not act on the target register state, since the first $2L-k$ qubits of the control register are in a product state and since $\operatorname{Tr}[|\phi\rangle \langle \psi|] = \langle \psi|\phi\rangle$ we get a Kronecker Delta δ_{ij}, therefore

$$\rho_k^{red} = \frac{1}{2^k} \sum_{i=0}^{2^k-1} |Y^i \bmod N\rangle \langle Y^i \bmod N| \tag{4.2}$$

We notice that the reduced density matrix is diagonal, so we analyze the sequence $\{Y^i \bmod N\}_i$: by definition of period, the infinite sequence is periodic with period r, so when we truncate the series to a finite sequence of length 2^k, we have q complete repetitions of the series and $c = 2^k - qr$ elements repeated an additional time, where

$$q = \operatorname{floor}(\frac{2^k}{r})$$

with floor : $x \to f, x \in \mathbb{R}, f \in \mathbb{N} : f \leq x < f+1$.
This means that in the sum in Equation 4.2 we have $r-c$ distinct terms repeated q times and c terms repeated $q+1$ times. By including the factor 2^{-k} we get the entanglement entropy

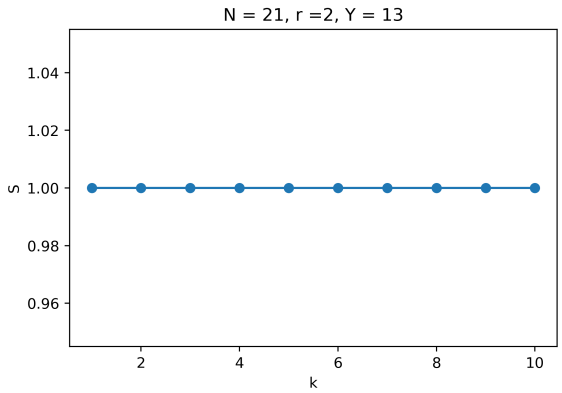
$$S_{k,r} = -(r-c)\frac{q}{2^k}\log_2\left(\frac{q}{2^k}\right) - c\frac{q+1}{2^k}\log_2\left(\frac{q+1}{2^k}\right) \leq \log_2(r)$$

where the last inequality comes from $\operatorname{rank}(\rho_k^{red}) \leq r$.
Since $\operatorname{rank}(\rho_k^{red}) = r \ \forall k : 2^k - 1 \geq r$ and the order is in general $r \in O(N)$, we find the entanglement entropy to be $O(N)$, or equivalently $O(2^L)$. Thus, we have found a particular bipartition for which the entanglement entropy is exponential in the number of qubits.
Therefore, Shor's algorithm can provide an exponential speedup.

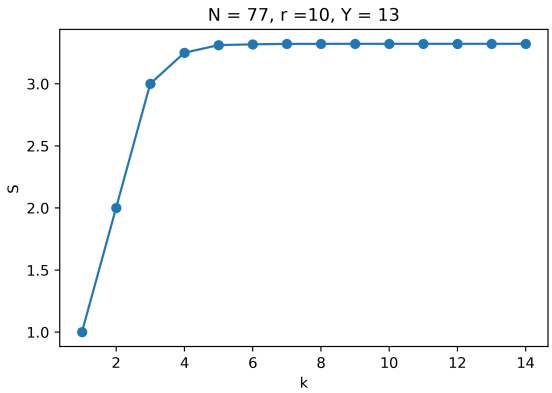
4.2 Simulation results

4.2.1 $S(k|Y,N)$ behaviour

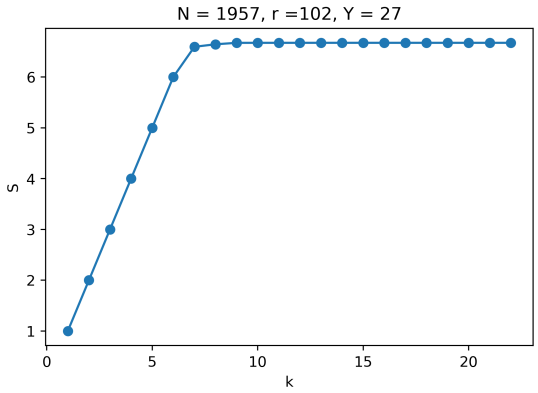
We first observe the evolution of the entanglement entropy as a function of the computational step.



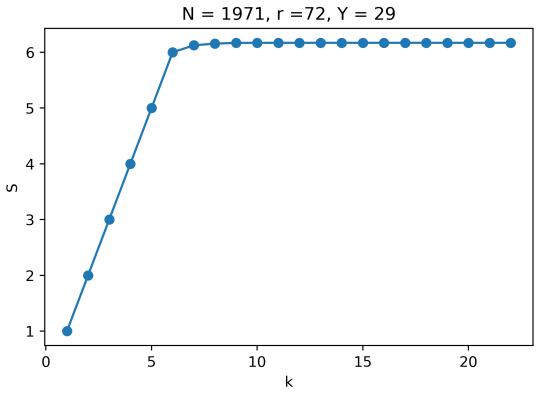
(a) $S(k)$ for $N = 21, Y = 13, r = 2$



(b) $S(k)$ for $N = 77, r = 10, Y = 13$



(c) $S(k)$ for $N = 1957, r = 102, Y = 27$



(d) $S(k)$ for $N = 1971, r = 72, Y = 29$

Figure 4.1: $S(k)$ for different values of N, r, Y

As we can see in Figure 4.1, the entropy between the registers $S(k)$ grows from $S(1) = 1$ to its maximum value $S(2L) \approx \log_2(r)$. Furthermore, the entropy increases almost linearly for computational steps $k \in [1, \log_2(r)]$, after approximately $\log_2(r)$ steps the entropy remains almost constant.

We can therefore state that the dynamics of the entropy between registers $S(k|N, r, Y)$ depends only on the order $r(Y, N)$.

4.2.2 $S(N|k = 2L)$ behaviour

Since we are free to choose how to perform the modular exponentiation part of the circuit, the state $|\psi_{2L}\rangle$ is the only one which is always shared among all implementations of the algorithm for $k \in \{1, .., 2L\}$.
We show a scatter plot of entanglement entropy S as a function of N at the end of the modular exponentiation. Each point represents the entanglement entropy for a pair (Y, N) with Y a coprime of N. The Y parameter is not considered here in order to study an average behaviour depending only on the (N,S) pair.

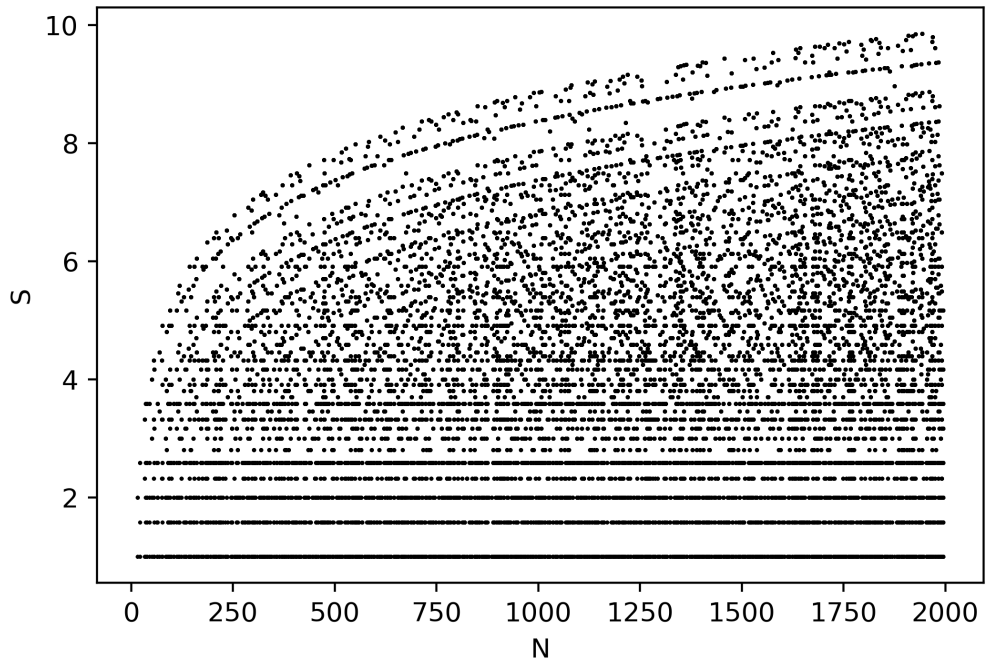


Figure 4.2: $S(N)$ at the end of the modular exponentiation, all Y coprimes of each N are represented

In Figure 4.3, we show a heatmap obtained from the same data. It shows concentration of the entanglement entropy across the (N, S) plane, namely $p_N(S)$.

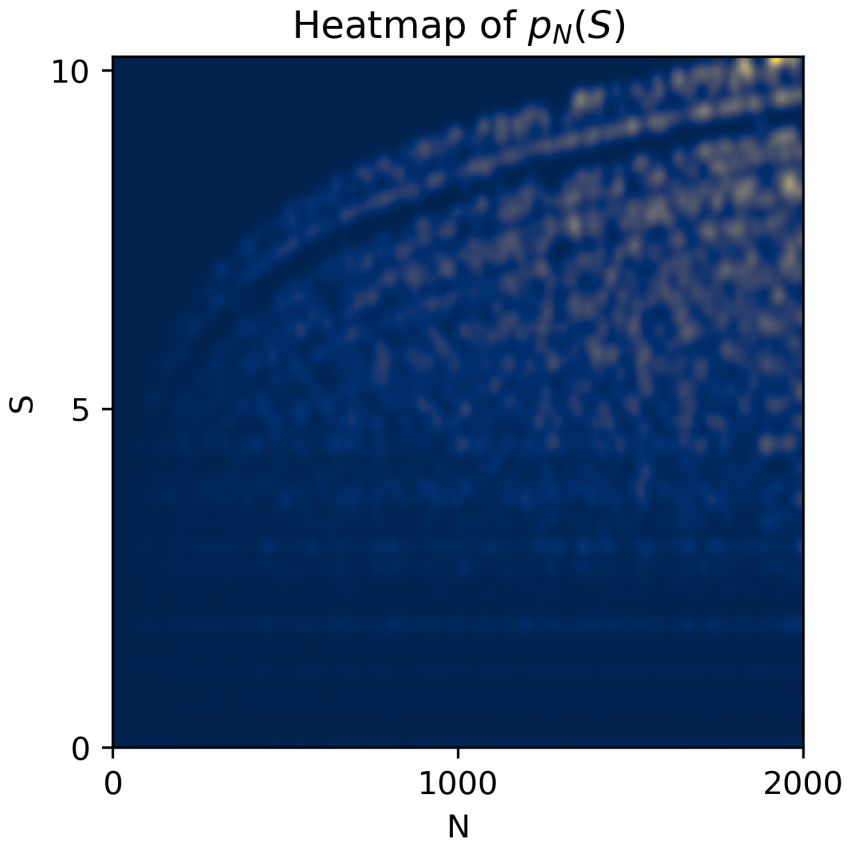


Figure 4.3: Heatmap of $p_N(S)$, obtained by considering a mesh in the (N, S) plane. The color represents the value of $p_N(S)$ inside a rectangle. Dark blue is zero, yellow represents high probability.

We can observe the entanglement entropy probability increases around a logarithmic upper bound. As N grows, $p_N(S)$ seems to concentrate more and more around this logarithmic convex hull.
We define the convex hull of the data.

$$\hat{S}(N) = \max_{Y} S(Y, N)$$

In Figure 4.4 we can see that the convex hull data points fluctuate between different logarithmic behaviours.
In Figure 4.5 we show the convex hull data points in semi-logarithmic scale. We

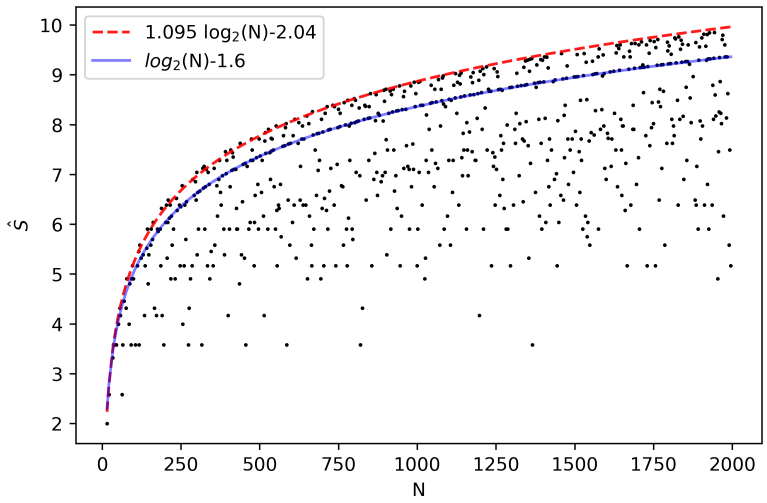


Figure 4.4: Convex hull $\hat{S}_N$ at the end of the modular exponentiation

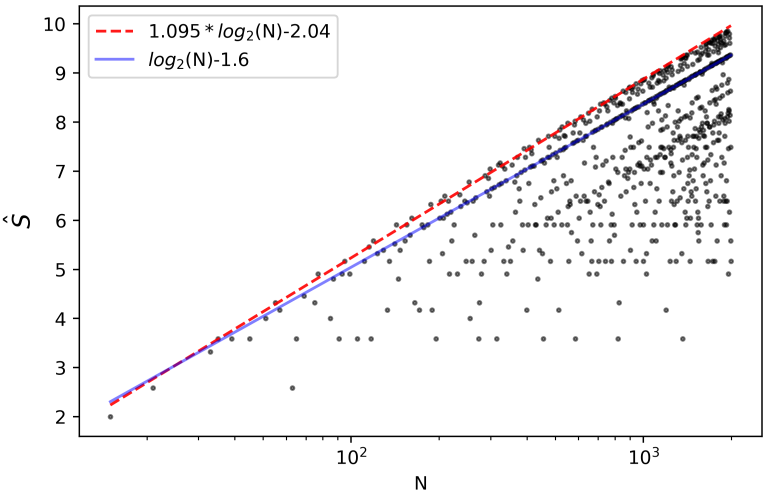


Figure 4.5: Convex hull $\hat{S}_N$ at the end of the modular exponentiation in semi-logarithmic scale

estimate a logarithmic upper bound and the most populated curve. In red, we estimated a logarithmic upper bound for the convex hull :

$$\hat{S} \leq 1.095 \log_2(N) - 2.04$$

.
In blue, we estimated the most populated line and verified it follows a logarithmic curve $y = \log_2(N) - 1.6$.
The curves in Figure 4.4 are used to describe some qualitative property of the data, but they had to be estimated from selected data points.
We now want to describe some property of the entropy. and therefore of the order r, in a more rigorous way.
Let us define $\mu_S(N)$ as the average entropy of $S(N)$ overall all Y coprimes of N, namely

$$\mu_S(N) = \underset{Y:GCD(Y,N)==1}{\text{avg}} S(Y,N) \qquad \forall Y \in \{2,..,N-1\}$$

We divide the domain $\{15,..,2000\}$ into distinct subsets N_i of equal length *step*,i.e.

$$N_i = \{15 + (i \cdot step),..,15 + (i+1) \cdot step - 1\}$$

Then we consider $\mu_S(N, step)$ as $\mu_S(N)$ averaged on all $N \in N_i$:

$$\mu_S(N, step) = \{\underset{N \in N_i}{\text{avg}} \mu_S(N) \ \forall i\}$$

We show this quantity as a function of N using semi-logarithmic scale in Figures 4.6 to 4.9. The blue line shows the estimated linear behaviour and the light blue area represents points in the data space which differs from the estimated behaviour for one standard deviation.
As we can see, the behaviour is similar for all values of the *step* variable: $\mu_S(N, step)$ follows a linear trend with respect to $\log_2(N)$ and the variance of the data grows with N. Furthermore, the parameters of the linear fit are not strongly correlated with the *step* hyper-parameter.
We can conclude

$$E[\mu_S(N, step)] \approx 0.8 \log_2(N) - 1.6 \quad \forall step$$

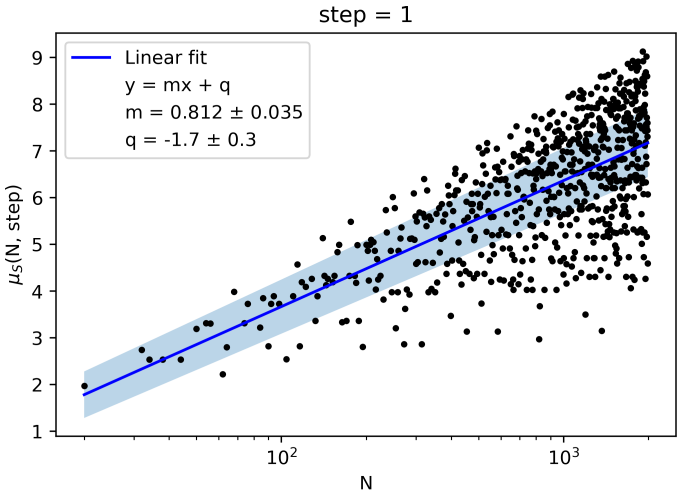


Figure 4.6: $\mu_S(N, step)$ for $step = 1$

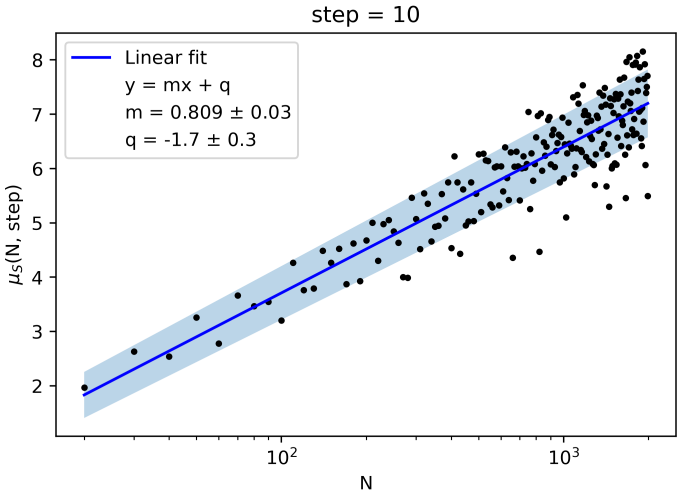


Figure 4.7: $\mu_S(N, step)$ for $step = 10$

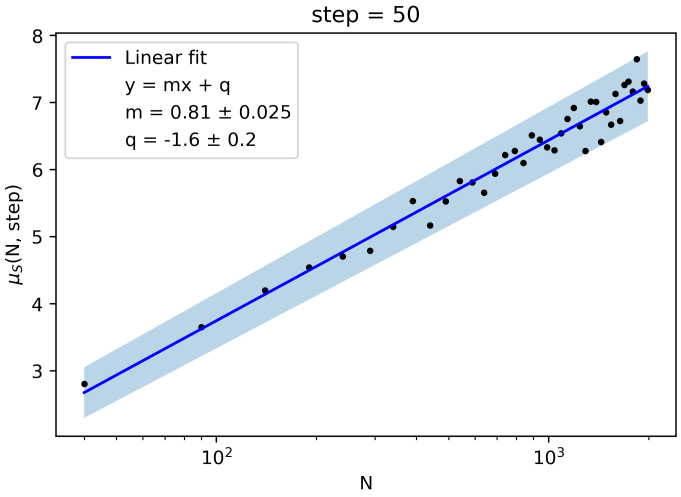


Figure 4.8: $\mu_S(N, step)$ for $step = 50$

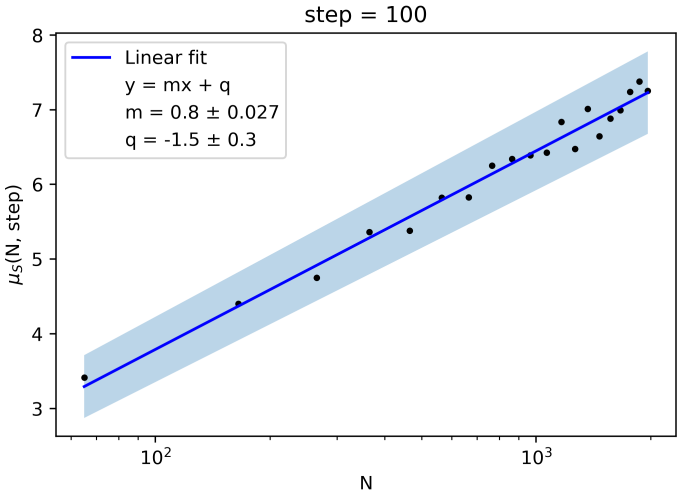


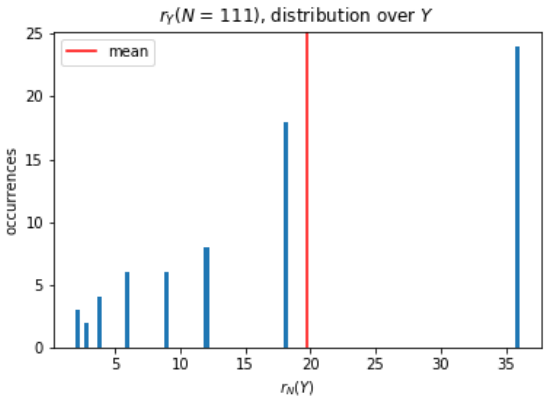
Figure 4.9: $\mu_S(N, step)$ for $step = 100$

4.2.3 Behaviour of the order

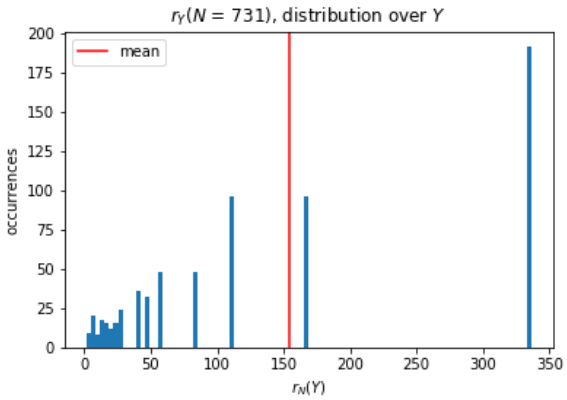
Since, as we have seen, the entropy $S(Y, N, k = 2L) \approx \log_2(r_{Y,N})$, we can transform Figures 4.6 to 4.9 into graphs of average $\log_2(r)$ as a function of N. Thus, if we define $\hat{r} = 2^{\mu_S(N,step)}$, we have observed

$$E[\hat{r}] \approx C\, N^m$$
$$C = 0.4 \pm 0.2 \quad m = 0.8 \pm 0.1$$

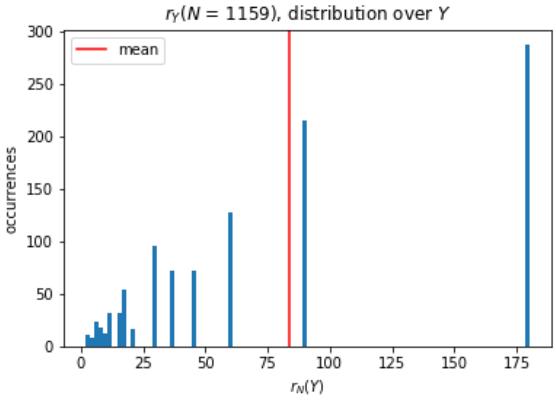
Since there is this a direct link between order and entropy, we now look at the order distribution for fixed N in order to gather some other information about the entropy distribution.
Can the estimate on $\hat{r}$ be used for estimating the order (and therefore, entanglement entropy) of a specific (Y, N) pair? Is the distribution of $r(Y|N) = r_N(Y)$ peaked on the mean value, for fixed N?
As we can see from Figure 4.10, the behaviour of $r_N(Y)$ is very similar across different values of N. The maximum value $\tilde{r}_N$ is also the mode, but this peak is very isolated from the rest of the values.
So for fixed N, most of the probability is concentrated in a few large values of $r_N(Y)$, while a larger set of smaller values have fewer occurrences each.
We therefore notice that $r_N(Y)$ as a distribution is peaked on the maximum order $\tilde{r}_N = \max_Y r_N(Y)$, but this distribution is not regular enough to make a proper inference around the mean value.



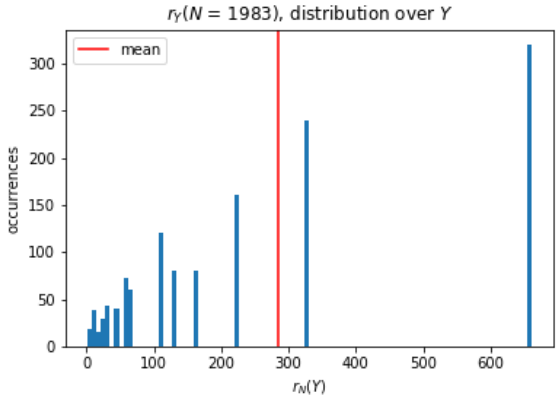
(a) $r_N(Y)$ as a distribution for $N = 111$



(b) $r_N(Y)$ as a distribution for $N = 731$



(c) $r_N(Y)$ as a distribution or $N = 1159$



(d) $r_N(Y)$ as a distribution for $N = 1983$

Figure 4.10: Distribution $r_N(Y)$ for various values of N

5. Arbitrary bipartition

We now consider a generic balanced bipartition of the $3L$ qubits, which does not only divide the two registers. For this generic case, the entanglement entropy can't be expressed analytically, as we did in the previous Chapter.
The lack of analytical results forces us to simulate the quantum algorithm in order to compute $\Lambda(\rho)$ numerically. Furthermore in this situation, unlike in the control-target bipartition, the IQFT operation will change the entanglement entropy of our state, so we need to simulate this transformation too.
In this Chapter, we will first introduce the concepts and ideas used to simulate Shor's algorithm on a classical computer. We will then discuss in detail different implementations, approaches and their features.

5.1 Entropy-preserving projection

Consider the state ρ^k of the whole $3L$ dimensional system at the k-th computational step of Shor's algorithm. Define I as the set of untouched qubits: for $k \in \{1, .., 2L\}$

$$I = \{0, 2, ..., 2L - k\}$$

The set I represents the set of qubits in control register which have not been used as control qubits by the modular exponentiation operation at step k. Each of the qubits in I is in the pure state $\rho_+ = |+\rangle \langle +|$.
From Equation 4.1, we can decompose the density matrix ρ^kinto factors as

$$\rho^k = (\bigotimes_{i \in I} \rho_i^+) \otimes \rho_{L+k}$$

where ρ_{L+k} is an operator acting on $\mathbb{C}^{2^{(L+k)}}$.
If a density matrix is a product state, i.e. $\rho = \rho_1 \otimes \rho_2$, then

$$S(\rho_1 \otimes \rho_2) = S(\rho) = S(\rho_1) + S(\rho_2) \tag{5.1}$$

Notice the similarity between this result in quantum information theory and the classical case for independent random variables:

$$X \perp Y \iff H(X,Y) = H(X) + H(Y) \iff I(X,Y) = 0$$

Theorem: $S(\rho_1 \otimes \rho_2) = S(\rho) = S(\rho_1) + S(\rho_2)$

Proof.
Consider the spectrum of $\rho_1 = \{\lambda_i^{(1)}\}_i = \Lambda(\rho_1)$ and same for $\{\lambda_j^{(2)}\}_j$. Trace normalization is ensured by $\sum_i \lambda_i^{(1)} = \sum_j \lambda_j^{(2)} = 1$. Then

$$S(\rho) = \sum_{i,j} \lambda_i^{(1)} \lambda_j^{(2)} \log(\lambda_i^{(1)} \lambda_j^{(2)}) = \sum_{i,j} \lambda_i^{(1)} \lambda_j^{(2)} \log(\lambda_i^{(1)}) + \sum_{i,j} \lambda_i^{(1)} \lambda_j^{(2)} \log(\lambda_j^{(2)})$$

$$= \sum_i \lambda_i^{(1)} \log(\lambda_i^{(1)}) + \sum_j \lambda_j^{(2)} \log(\lambda_j^{(2)}) = S(\rho_1) + S(\rho_2)$$

Since each ρ_i^+ is a pure state, Equation 5.1 implies $S(\rho^k) = S(\rho_{L+k})$. Therefore we can compute the entanglement entropy directly on the projected density matrix ρ_{L+k}, effectively lowering the dimension of the problem without any loss of precision $\forall k < 2L$.
We can then effectively substitute to the state in Equation 4.1 **an equivalent state** of only $k + L$ qubits $|\psi_k\rangle$.

$$|\psi_k\rangle = \frac{1}{2^{k/2}} \sum_{l_{2L-k+1}=0}^{1} \cdots \sum_{l_{2L}=0}^{1} |l_{2L-k+1} \dots l_{2L}\rangle \, |Y^{\sum_{n=0}^{k-1} l_{2L-n} 2^n} \bmod N\rangle \tag{5.2}$$

Notice that the projected state has sparsity $\sigma(|\psi_k\rangle)$

$$\sigma(|\psi_k\rangle) = \frac{2^{k+L} - 2^k}{2^{k+L}} = 1 - 2^{-L}$$

So the projected state is highly sparse for $L \geq 4$.

5.2 State initialization and modular exponentiation

Using the equation 4.1 and the projection discussed in the previous section, we can write the state of the system at each computational step $k \in [1, 2L]$.
As discussed, the state is highly sparse and it is represented by

$$|\psi_k\rangle = \frac{1}{\sqrt{2^k}} \sum_{i=0}^{2^k-1} |i\rangle \otimes |Y^i \bmod N\rangle$$

$$= \frac{1}{\sqrt{2^k}} \sum_{i=0}^{2^k-1} \begin{matrix} 0\{ \\ 1\{ \\ \\ i\{ \\ \\ \\ \\ \end{matrix} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \} \quad Y^i \bmod N$$

Then, the state nonzero entries are located at indices

$$J_k = \{j \cdot 2^L + Y^j \bmod N, j \ \in [0, 2^k - 1]\}$$

Let $|\psi\rangle[j]$ be the j-th entry of the column vector $|\psi\rangle$

$$|\psi_k\rangle[j] = 2^{-k/2} \quad \forall j \in J_k \quad \forall k \in [1, 2L] \tag{5.3}$$

Computing the element of J_k using a straightforward approach can easily lead to overflow and repeating a lot of computations.
In order to compute the element of the set J_k in a computationally more efficient way, we use a distributive property of modular multiplication: let $a, b, N \in \mathbb{N}$ then

$$(a \cdot b) \bmod N = [(a \bmod N) \cdot (b \bmod N)] \bmod N$$

Applying to modular exponentiation, we get a recursive algorithm

$$Y^n \bmod N = [(Y^{n-1} \bmod N) \cdot (Y \bmod N)] \bmod N$$

This formulation reduces the computational time by an order of magnitude and prevents overflow, since each operation involves number which are $O(N^2)$.
More complex and parallel algorithms for modular exponentiation have been developed, due to the central role of modular exponentiation in classical cryptography. Still the plain recursive algorithm reduced the computational time so that optimizing this part of the code would be useless with respect to other bottlenecks for our computation.

5.3 W formulation for reduced density matrices

In the straightforward approach, computing a reduced density matrix ρ_{red} would require to store the whole density matrix ρ and apply the partial trace operation on it. Since our vector state is $\mathrm{O}(2^{3L})$ dimensional, the complete state ρ would need $\mathrm{O}(2^{6L})$ complex entries. If $L = 8$, using complex amplitudes of at least 48 bytes each (24 for each float representing real and imaginary part), storing the whole ρ matrix would require an unfeasible amount of memory. Obviously, the cost of storing ρ can be reduced by exploiting its properties.
We propose a way to compute directly the reduced density matrix for a pure state using at most 3 matrices of $O(2^{3L})$ complex entries each, which is equivalent to the cost of storing 3 pure states.
Let us consider a system of n qubits, then fix $T = \{|t\rangle\}_i$, $i \in I$ a orthonormal basis of $\mathbb{C}^{\mathbb{N}}$, with $N = 2^n$. Let $|\psi\rangle = \sum_t c_t |t\rangle \in \mathbb{C}^{\mathbb{N}}$ then $\rho = \sum_{i,j \in I} c_i c_j^* |i\rangle \langle j|$.
Fix a set B of n_b qubits as environment and A as the considered qubits, then $\rho_A = \mathrm{Tr}_B(\rho)$. Let $Z_B = \{|z_b\rangle\}_z$ be the basis related to the environment. Let $\{|v_a\rangle\}_a$ be the basis for the state of the system A. Considering a suitable qubits ordering, then $\forall\, |i\rangle \in I\ \exists!(|v_a\rangle, |z_b\rangle)$ pair such that

$$|i\rangle = |v_a\rangle \otimes |z_b\rangle$$

Then

$$|\psi\rangle = \sum_{a,b} c_{a,b} |v_a\rangle \otimes |z_b\rangle$$

$$\rho = \sum_{a,b,a',b'} c_{a,b}\ c^*_{a',b'} |v_a, z_b\rangle \langle v_{a'}, z_{b'}|$$

$$\begin{aligned}\mathrm{Tr}_B(\rho) &= \sum_{b''} \sum_{a,b,a',b'} c_{a,b}\ c^*_{a',b'} \left\langle z_{b''} | v_a, z_b \right\rangle \left\langle v_{a'}, z_{b'} | z_{b''} \right\rangle \\ &= \sum_{b,a,a'} c_{a,b}\ c^*_{a',b} \left| v_a \right\rangle \left\langle v_{a'} \right| = \rho_A\end{aligned} \tag{5.4}$$

Then

$$\rho_A[a, a'] = \langle a | \rho | a' \rangle = \sum_b c_{a,b}\ c^*_{a',b}$$

.

Define a matrix W of dimension $(2^{n_a} \times 2^{n_b})$ as a remapping of the state $|\psi\rangle$ such that

$$W[a, b] = c_{a,b}$$

then

$$W\, W^\dagger[a, a'] = \sum_b W[a, b] W^\dagger[b, a'] = \sum_b c_{a,b}\, c^*_{b,a'} = \rho_A[a, a']$$

which implies

$$W\, W^\dagger = \rho_A \tag{5.5}$$

This formulation allows us to compute the reduced density matrix having a memory cost of $O(2^{3L})$, which is the same asymptotic cost of storing the pure state.[2]

If we consider the Equation 5.5, we can see that computing the eigenvalues of ρ can be achieved by using the singular value decomposition of the W matrix. Let $\Sigma(W)$ be the set of singular values of W, then

$$\Lambda(\rho) = \{\sigma_w^2 \text{ s.t. } \sigma_w \in \Sigma(W)\}$$

This relation can be really useful, since it let us compute the spectral decomposition of ρ without having to compute the final reduced density matrix. Furthermore, we can try using an approximate SVD for high dimensional systems in order to compute an approximation of $\Lambda(\rho)$.

On the other hand, if we compute the final reduced density matrix and we consider negligible some numerical error, ρ can be considered Hermitian. This property allows us to use some optimized algorithms if we compute the spectrum of ρ directly. No such general property can be stated for the W matrix, so in general we cannot use specialized methods in order to accelerate the SVD decomposition.

[2]Overwriting matrices, storing only upper or lower diagonal for the Hermitian matrix ρ and breaking the dot product into smaller dot products can further decrease the memory cost, but it would still be $O(2^{3L})$.

5.3.1 Examples of W

As an example, consider three qubits $Q0$, $Q1$, $Q2$ and their generic state

$$|\psi\rangle = \begin{bmatrix} \alpha_{000} \\ \alpha_{001} \\ \vdots \\ \alpha_{111} \end{bmatrix}$$

Then, in order to compute ρ_{Q2}, we have a (2×4) matrix W as

$$W = \begin{bmatrix} \alpha_{000}, & \alpha_{010}, & \alpha_{100}, & \alpha_{110} \\ \alpha_{001}, & \alpha_{011}, & \alpha_{101}, & \alpha_{111} \end{bmatrix} \tag{5.6}$$

Then $W\,W^\dagger = \rho_2$ is

$$\rho_{Q2} = \begin{bmatrix} |\alpha_{000}|^2 + |\alpha_{010}|^2 + |\alpha_{100}|^2 + |\alpha_{110}|^2, & \alpha_{000}\,\alpha_{001}^* + \alpha_{010}\,\alpha_{011}^* + \alpha_{100}\,\alpha_{101}^* + \alpha_{110}\,\alpha_{111}^* \\ \alpha_{000}^*\,\alpha_{001} + \alpha_{010}^*\,\alpha_{011} + \alpha_{100}^*\,\alpha_{101} + \alpha_{110}^*\,\alpha_{111}, & |\alpha_{001}|^2 + |\alpha_{011}|^2 + |\alpha_{101}|^2 + |\alpha_{111}|^2 \end{bmatrix}$$

If we choose instead to trace away the qubit $Q1$ and consider only the qubits $Q0$ and $Q2$, we have a (4×2) matrix W as

$$W = \begin{bmatrix} \alpha_{000} & \alpha_{010} \\ \alpha_{001} & \alpha_{011} \\ \alpha_{100} & \alpha_{110} \\ \alpha_{101} & \alpha_{111} \end{bmatrix} \tag{5.7}$$

Then $W\,W^\dagger = \rho_{Q0.Q2} = \rho$ is

$$\rho = \begin{bmatrix} |\alpha_{000}|^2 + |\alpha_{010}|^2, & \alpha_{000}\,\alpha_{001}^* + \alpha_{010}\,\alpha_{011}^*, & \alpha_{000}\,\alpha_{100}^* + \alpha_{010}\,\alpha_{110}^*, & \alpha_{000}\,\alpha_{101}^* + \alpha_{010}\,\alpha_{111}^* \\ \alpha_{001}\,\alpha_{000}^* + \alpha_{011}\,\alpha_{010}^*, & |\alpha_{001}|^2 + |\alpha_{011}|^2 & \alpha_{001}\,\alpha_{100}^* + \alpha_{011}\,\alpha_{110}^*, & \alpha_{001}\,\alpha_{101}^* + \alpha_{011}\,\alpha_{111}^* \\ \alpha_{100}\,\alpha_{000}^* + \alpha_{110}\,\alpha_{010}^*, & \alpha_{100}\,\alpha_{001}^* + \alpha_{110}\,\alpha_{011}^*, & |\alpha_{100}|^2 + |\alpha_{110}|^2 & \alpha_{100}\,\alpha_{101}^* + \alpha_{110}\,\alpha_{111}^* \\ \alpha_{101}\,\alpha_{000}^* + \alpha_{111}\,\alpha_{010}^*, & \alpha_{101}\,\alpha_{001}^* + \alpha_{111}\,\alpha_{011}^*, & \alpha_{101}\,\alpha_{100}^* + \alpha_{111}\,\alpha_{110}^*, & |\alpha_{101}|^2 + |\alpha_{111}|^2 \end{bmatrix}$$

5.3.2 W construction in parallel

W has another useful property: its construction can be easily made parallel. Here we will show how.
We first look at some example in order to understand the structure of the problem: let us consider a 3 qubit state

$$|\psi\rangle = \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_7 \end{bmatrix}$$

If we consider the reduced density matrix for the last qubits we get the W matrix in Equation 5.6. We can define an integer valued matrix M with same shape as W, which we will call mapping matrix, s.t.

$$W[i,j] = \alpha_{M[i,j]}$$

We get

$$M = \begin{bmatrix} 0 & 2 & 4 & 6 \\ 1 & 3 & 5 & 7 \end{bmatrix}$$

When we want to trace away the middle qubit, we get the W matrix in Equation 5.7 and therefore a mapping matrix M

$$M = \begin{bmatrix} 0 & 2 \\ 1 & 3 \\ 4 & 6 \\ 5 & 7 \end{bmatrix}$$

By construction, the value of $M[i,j]$ in binary representation has the preserved qubits in state i (as a binary string) and the environment qubits in state j. Furthermore, we notice that for fixed j, the gap $g_j = M[i,j+1] - M[i,j]$ is constant with respect to i.

Then, in order to generate the whole mapping matrix M for a general set of chosen qubits $C = \{c_0, .., c_n\}$ out of the total 3L qubits system, we can compute the first column $M[i,0]\,\forall i \in \{0, .., 2^n - 1\}$ and $g_j\,\forall j \in [0, 2^{3L-n-1}]$. This brings the computation needed to generate M from $O(2^{3L})$ to $O(2^m)$ where $m = \min(n, 3L - n)$.

Since the computation of gaps and first column is independent, we can perform this in parallel. Even the computation of each element of the gap and first column can be carried out independently.

In Figure 5.1, we show the average time needed to compute the entanglement entropy for 1000 bipartitions with the traditional (serial) method and by using a numba accelerated implementation (parallel) on 40 cores.

If we consider the last and most computationally expensive step, $k = 17$, we have a construction time $t_c = 0.34 \pm 0.07\ s$ and a GPU-accelerated eigenvalue decomposition

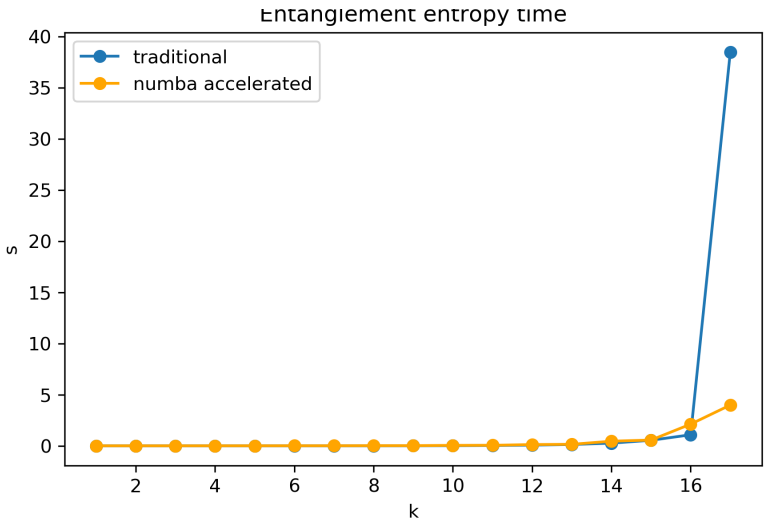


Figure 5.1: Scaling for serial and parallel implementation, L=8

time for Hermitian matrices $t_e = 3.65 \pm 0.08\ s$. The bottleneck of the algorithm now resides in the spectral decomposition of the reduced density matrix $\rho_{red} = WW^\dagger$.
Remark: The computation of the whole M matrix is often efficient since it can be obtained from a few computations. When $|\psi\rangle$ is highly sparse though, it can be more efficient to compute each $M[i,j]$ in parallel for each nonzero entry of $|\psi\rangle$: if a state has N nonzero entries, we can still perform the whole N computations in parallel and fill the W matrix with zero entries, but the complexity would be limited by the number of the nonzero entries. So the computation would be $O(N)$ instead of $O(2^m)$. Whether or not this is beneficial depends on the size of the chosen qubits set C and the sparsity of $|\psi\rangle$.

5.4 State sparsity: computational and memory trade-off

We first introduce some notation. Let $n_{\neq 0}(A)$ be the number of nonzero entries of A. For fixed L, we denote by the computational step $k = 2L + 1$ the application of the IQFT on the control register.
We know the state at the end of modular exponentiation has $n_{\neq 0}(|\psi_{2L}\rangle) = 2^{2L}$ out of 2^{3L} complex coefficients. So it would be memory efficient to use a sparse representation in order to store them. Even though the number of nonzero entries is always the same, the distribution of this entries in the 2^{3L} - dimensional vector depends on the (Y, N) pair.
Defining the sparsity of a (m x n) matrix A as

$$\sigma(A) = 1 - \frac{n_{\neq 0}(A)}{mn} \tag{5.8}$$

We have $\sigma(|\psi_k\rangle) = 1 - 2^{-L} \quad \forall k \in [1, 2L]$.
At the the end of the IQFT, the number of nonzero elements depends on the distribution of the nonzero values after the modular exponentiation, therefore it's determined by the (Y, N) pair.
Let $i \in [0, 2^{3L} - 1]$ and $j = i \bmod 2^L$, after the IQFT

$$|\psi_{2L+1}\rangle [i] = 0 \iff |\psi_{2L}\rangle [j + m \cdot 2^L] = 0 \ \forall m \in [0, 2^{2L} - 1]$$

Formally, define slices of the vector state into 2^L dimensional parts $I_0, ..., I_{2^{2L}-1}$, where

$$I_j = |\psi_{2L+1}\rangle [\, j \cdot 2^L, .., (j + 1) \cdot 2^L - 1 \,]$$

Each I_j has exactly one nonzero element, whose index in the I_j vector is

$$i_j^\star = Y^j \bmod N$$

If the set of all i_j^* covers all the set of natural numbers in $[0, 2^L - 1]$, then the state after IQFT is completely dense, formally

$$\{i_j^\star\}_j \cap [0, 2^L - 1] = [0, 2^L - 1] \implies n_{\neq 0}(|\psi_{2L+1}\rangle) = 2^{3L}$$

So for a generic pair (Y, N), we must be able to store the whole 2^{3L} dimensional complex vector in order to study the state after IQFT.
Since in general we are required to store a completely dense vector after the IQFT operation, we can freely choose whether to use sparse or dense representation of the array.

Since dense singular value decomposition (SVD) and eigenvalues algorithms are much faster than sparse ones, we decided to use dense representation.

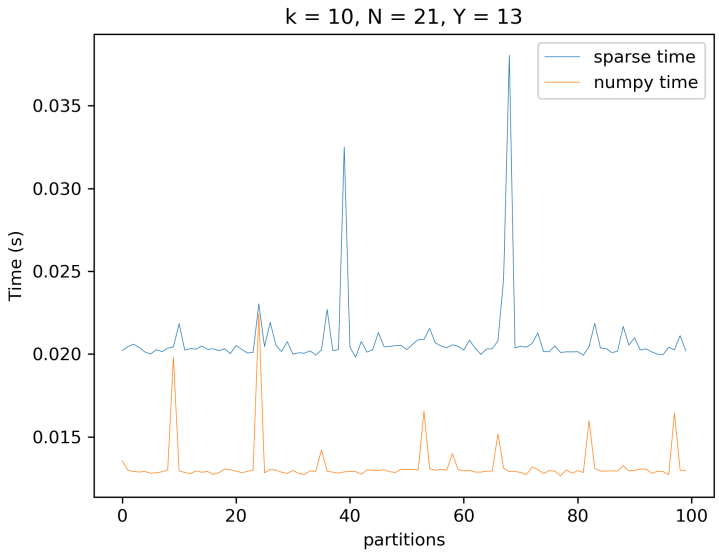


Figure 5.2: An example of comparison between sparse and dense methods: scipy sparse SVD method vs numpy SVD

We see in Figure 5.2 that dense representation leads to a huge speedup in the eigenvalue decomposition, needed in order to compute the entanglement entropy.

5.5 IQFT Operation

The IQFT operation is performed on the control register and therefore it acts on the whole state as

$$Q = M \otimes \mathbb{1}_{2^L} \tag{5.9}$$

with

$$M_{j,k} = 2^{-3L/2}\,\omega^{-j\cdot k} \quad \forall j,k \in [0, 2^{2L}-1], \quad \omega = e^{i\pi 2^{2L-1}} \tag{5.10}$$

This matrix, if represented as a dense operator,consists of 2^{6L} entries and it is the memory bottleneck of the computation.
We first searched libraries implementing this transformation and found Qiskit [20], QuTiP [21] and Quantum++ [22]. Qiskit and QuTiP cannot exploit multicore architectures and only exploit shared memory (openMP), while Quantum ++ is an MPI optimized library in C++, but it lacks flexibility and documentation. All these libraries only support dense representation of states and gates.
We chose to use Qiskit framework as it is more flexible, has a clearer documentation and can be easily connected with external code. IQFT operation scales exponentially in time and memory, as we would expect.

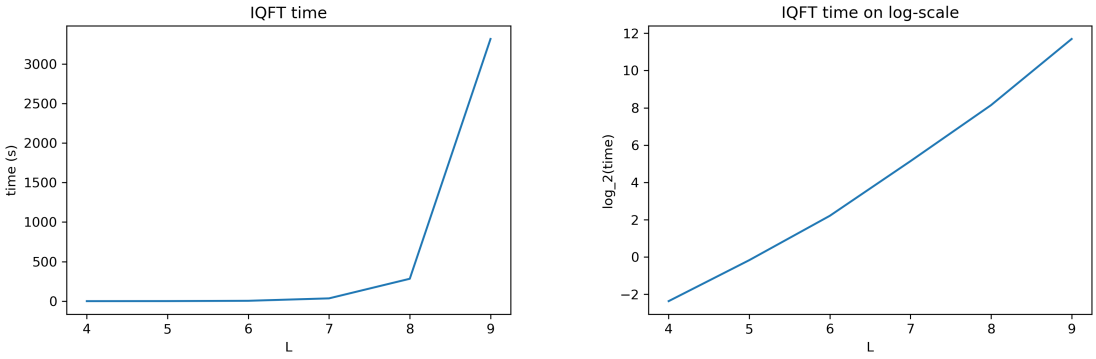


Figure 5.3: IQFT time scaling

We now want to determine whether the simulation bottleneck for execution time is represented by the IQFT transformation or the eigenvalue decomposition for entropy calculation. We first search for the most efficient entropy finding algorithm: we considered three different algorithms:

– qiskit modular: time needed to compute $S(|\psi_k\rangle)$ for 100 bipartitions $\forall k \in [1, 2L]$ using Qiskit data structures and methods.

– scipy modular: time needed to compute $S(|\psi_k\rangle)$ for 100 bipartitions $\forall k \in [1, 2L]$ using scipy.sparse data structures and methods. W restatement for computing reduced density matrix is implemented.

– numpy modular: time needed to compute $S(|\psi_k\rangle)$ for 100 bipartitions $\forall k \in [1, 2L]$ using numpy data structures and methods. W restatement for computing reduced density matrix is implemented.

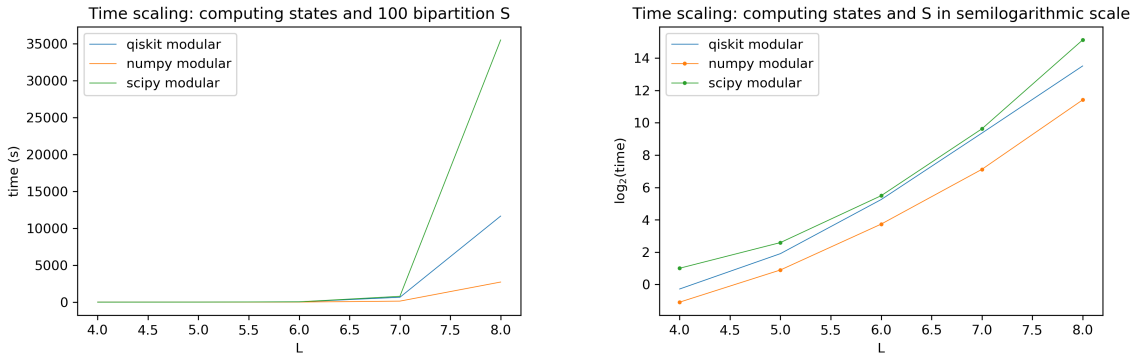


Figure 5.4: Scaling using different implementations

Figure 5.4 shows the comparison and the different scaling of these approaches.
The sparse implementation is by far the worst performing. The time taken by a straight-forward numpy implementation is always better than the Qiskit one.
We now stress that the IQFT operation will be performed only once on the state $|\psi_{2L}\rangle$. On the other hand, the spectral decomposition will be iterated for multiple bipartitions for each single computational step.
In Figure 5.5 we show the ratio between the time needed to compute the IQFT on $|\psi_{2L}\rangle$ using Qiskit and the time needed to compute the entanglement entropy using numpy algorithms with W restatement. We compute the entropy after the last computational step, we consider 100 bipartitions as benchmark and plot against L.
Since we will scale up to the number of bipartitions by at least an order of magnitude, we find that IQFT computation time will account only for a small percentage of the computational time, so there's no need to look for and test other libraries, nor designing an optimized IQFT algorithm.

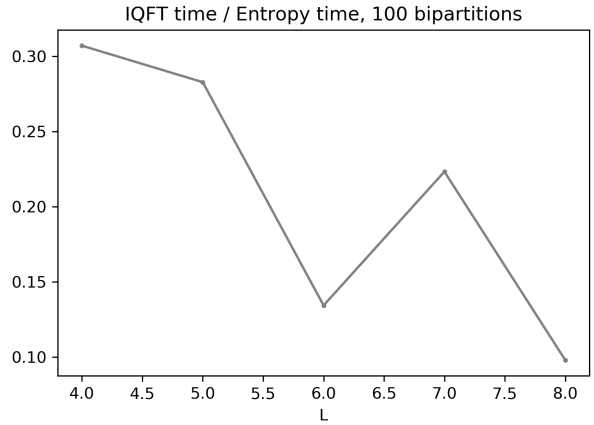


Figure 5.5: Ratio between time needed to compute the IQFT and the time to compute the entanglement entropy of 100 bipartitions

5.6 SVD approximation

As we saw in Figure 5.2 and Figure 5.4, sparse representation leads to a relevant time overhead when computing the singular values or, equivalently, the eigenvalues. This happens because the main library used for eigenvalue and svd decomposition of sparse matrices (here scipy.sparse wrapper), actually computes the eigenvalues of $\rho_{red} = W^T W$ using a FORTRAN 77 collection of subroutines called ARPACK and according to the documentation "The package is designed to compute a few eigenvalues and corresponding eigenvectors of a general n by n matrix A. It is most appropriate for large sparse matrices [...]. This software is based upon an algorithmic variant of the Arnoldi process called the Implicitly Restarted Arnoldi Method (IRAM)."

This algorithm, as many SVD and eigenvalue algorithms for sparse matrices, is designed to compute only a small subset of the whole spectrum, by iteratively computing the largest unknown singular value. A direct approach for singular value decomposition with large matrices would be computationally infeasible.

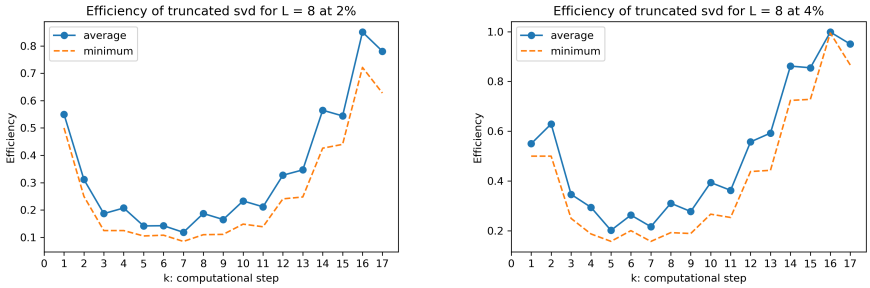
Furthermore, ARPACK is faster for high frequency of the spectrum, so the SVD is slower in finding the singular values closest to zero, which are the ones we are less interested about. We would like to use a partial SVD in order to approximate the entanglement entropy.

We tested whether the partial singular value decomposition provided by ARPACK for sparse matrices can be effectively applied to our problem.

Testing the algorithm scaling for our application, we can see that ARPACK's performance is equivalent to dense algorithms only when the fraction of computed eigenvalues $p_l \in [0.02, 0.04]$, for $p_l \geq 0.06$ the computational time needed by ARPACK makes it less effective than classical direct methods.
We define Efficiency $\text{Eff}(p_l, \rho_{red})$ as

$$\text{Eff}(p_l, \rho_{red}) = \frac{S(p_l, , \rho_{red})}{S(\rho_{red})}$$

where $S(p_l, , \rho_{red})$ is the Von Neumann Entropy of ρ_{red} computed using only a fraction p_l of the complete spectrum $\Lambda(\rho_{red})$ of ρ_{red}.
We computed the average and minimal efficiency among 100 random bipartitions for different values of p_l and for all $L \in [5, 9]$. Naturally, we noticed a steady increase in the average efficiency for fixed p_l as L grows. In figure 5.6 we plot the results for $L = 8$.



(a) Efficiency for $p_l = 0.02$

(b) Efficiency for $p_l = 0.04$

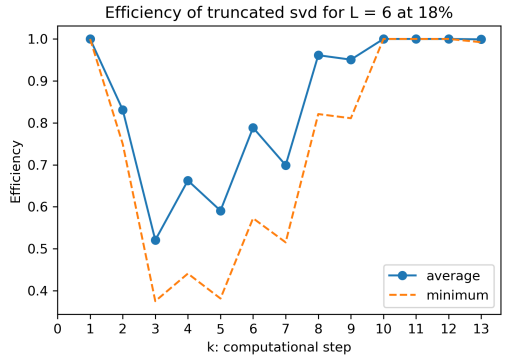
(c) Efficiency for $p_l = 0.18$

(d) Efficiency for $p_l = 0.20$

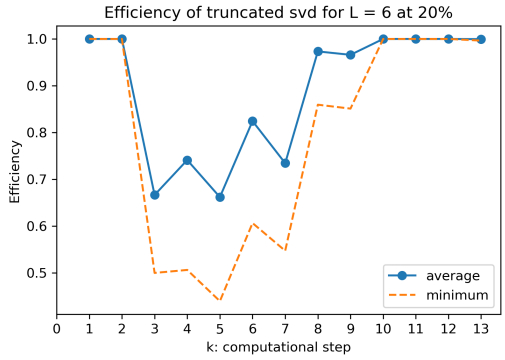
Figure 5.6: Efficiency for different percentages of eigenvalues, L = 8

Figure 5.6 shows that we cannot use a truncated spectrum for $L \leq 8$. It underlines

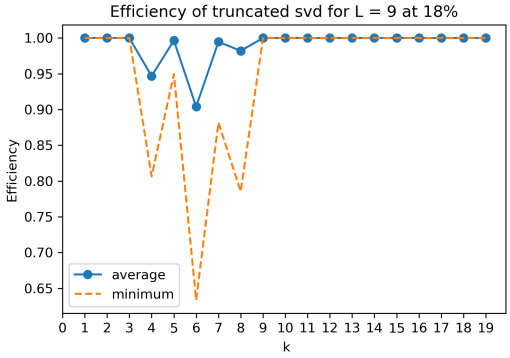
how our SVD cannot be efficiently truncated even by computing 20% of the singular values.
Despite the growth of average efficiency with L, we verified it still remains below 95% even for $L = 9$ and $p_l = 20\%$. Furthermore, even in these conditions, the minimum efficiency is still below 70%.



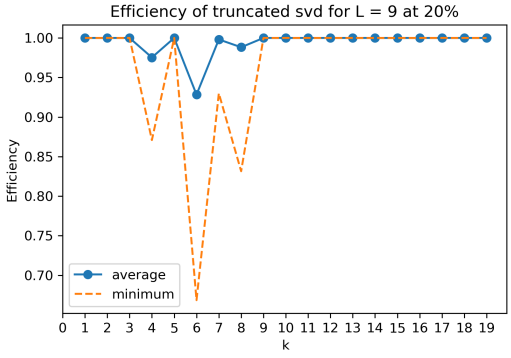
(a) Efficiency for L=6, $p_l = 0.18$



(b) Efficiency for L=6, $p_l = 0.20$



(c) Efficiency for L=9, $p_l = 0.18$



(d) Efficiency for L=9, $p_l = 0.20$

Figure 5.7: Efficiency for different values of L

We therefore cannot use partial SVD in order to approximate the entanglement entropy, not even for systems with a number of qubits $3L = 27$.
Other methods beside truncated SVD have been considered to approximate the entanglement entropy, for instance randomized SVD [23] and the use of approximating matrices. Due to the probabilistic nature of this class of methods, we decided to discard them: since our problem has no analytical solution and we have nearly no data to verify our results, we decided to explore deterministic algorithms only.

5.7 GPU implementation

We tried to use the GPU implementation of IQFT provided by Qiskit in order to accelerate the transformation, but as of now this implementation requires too much memory to be run on the 5GB NVIDIA Tesla K20.
Since the repeated eigenvalue decomposition is the bottleneck for our program, we tried to execute the eigenvalue computation on a GPU, using a CUDA wrapper. We first tested and benchamarked jax, an unofficial Google library, easily portable as it can be executed both on CPU or GPU.
Jax has an inferior precision with respect to Numpy and Scipy, but since we are interested in the distribution of the entropy, a precision of 6 decimal numbers is sufficient. The inferior precision gave us a speedup factor of 1.8 on average with respect to numpy. Due to a bug on the installer of the library though, the package could not be installed on the cluster.
We therefore changed wrapper and used cupy.
We moved the computations using dense matrices into the GPU using cupy. We have both a time limit due to the scheduler (12h) and a memory threshold due to the hardware specifics: our graphic card, a NVIDIA TESLA K20, has 5GB of memory. Due to memory limitations and dense representation, we can only use GPU eigenvalue decomposition on dense matrices for L ≤ 8.
For $L \geq 9$, we would need to use sparse GPU SVD algorithms, which are yet to be implemented both in cuSPARSE and cuSOLVE.
When dealing with sparse matrices and memory limitations we decided to use the SVD formulation since

- W has the same sparsity as the original state, which is $1 - 2^{-L}$ and increasing with L.
- $\rho_{red} = W\,W^{\dagger}$ have non-fixed sparsity and is in general more dense than W.

Since nor cuSPARSE or cuSOLVE provide a CUDA sparse SVD implementation, we provide the following solution using the eig function of cuSOLVE (wrapped in PyViennaCL, for example).
Construct A

$$A = \left[\begin{array}{c|c} 0 & W \\ \hline W^* & 0 \end{array}\right]$$

Now A is an Hermitian matrix whose eigenvalues $\Lambda(A) = \Sigma \cup (-\Sigma)$ with Σ the set of singular values of W.

The eig function in cuSOLVE for now cannot exploit hermitianity nor it supports the shift invert method for computing only the positive eigenvalues of A, but this formulation is such that A is an Hermitian matrix with only 2^{2L+1} nonzero elements.

6. Montecarlo simulation, validation and compatibility

One of the main problems in simulating a quantum algorithms consists in the exponential scaling in the linear dimension of ρ, which causes an exponential scaling in both memory and computational time.
For Shor's entropy dynamics at the k-th computational step, we discussed in the previous chapter how we can consider an equivalent system of $k + L$ qubits instead of a $3L$ qubits system, as in Equation 5.2 for $k \leq 2L$.
Furthermore, for fixed number of qubits $k+L$ even the number of (almost) balanced bipartitions $B(k, L)$ increases exponentially with the computational step, as

$$B(k, L) = \binom{k+L}{\text{floor}((k+L)/2)}$$

For $n = k + L$ even, $B(k, L)$ gives the central binomial coefficient CBC(n)[24]:

$$CBC(n) = \binom{2n}{n} \approx \frac{4^n}{\sqrt{\pi n}} \left(1 - \frac{1}{8n} + ...\right)$$

and for n odd, using the Stirling approximation

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

We obtain another exponential growth

$$\binom{n}{f} \approx \sqrt{\frac{n}{2\pi}} \left(\frac{n}{f}\right)^n \frac{1}{e(f+1)}, \qquad f = \text{floor}(n/2)$$

which, for $n \gg 1, n/f \approx 2$, gives

$$\binom{n}{f} \to C \frac{2^{n+1}}{e\sqrt{\pi n}}$$

Since we have no hypothesis on the distribution of the entanglement entropy in the bipartition space and for $k + L > 15$ we have no way of considering the whole space in a reasonable amount of time, we conduct a Montecarlo simulation in order to see whether we can approximate the entanglement entropy distributions only considering a subset of the bipartition space.
Data-driven methods for exploring the bipartition space cannot be easily applied: since the entanglement entropy is lockable, it's not trivial to define a useful distance D_B on the bipartition space.
Given the state ρ_{k+L}, since the entanglement entropy is symmetric with respect to complementary bipartitions, we can identify a bipartition b_1 by only the considered qubits: $b_1 = \{a_1, .., a_{(k+L)/2}\}, \quad a \in \{1, .., k + L\}$. Given two bipartitions b_1 and b_2, there's no evidence for existence of a distance D_B s.t.

$$D_B(b_1, b_2) < \delta \implies |S(\rho_{b1}) - S(\rho_{b2})| < \epsilon$$

where ρ_{b1} is the reduced density matrix of ρ_{k+L} obtained by tracing away the qubits not in b_1.
We would like to be able to approximate the entanglement entropy distribution only using a subset of the bipartition space. We selected a system which can be completely simulated with $N = 75$, $L = 7$, $k = 11$ and we formulated the hypothesis that subsets of the data yields compatible distributions. As we can see in Figure 6.1, considering only 35% of the data, the hypothesis holds.
If we look at the distribution after the IQFT decomposition, i.e. when the bipartition space has maximum dimension for fixed L, we still see convergence as in Figure 6.2. The same has been verified for $N = 255$, $k = 8$.
We then perform a Montecarlo simulation, in which we define convergence whenever both mean and variance of the entanglement entropy between steps change for less than a tolerance factor. We fixed this tolerance factor to 10^{-2}, resulting in a standard error on mean and variance of order 10^{-2} .Convergence is achieved in few Montecarlo steps and the results are compatibles the ones obtained for the complete bipartition space.
Since the sampling from the bipartition space is random, we verified that independent converged simulations for same N, Y, k gives the same distribution. We have done this in two ways:

i. Performing a two-sample Kolmogorov-Smirnov test on the data [25]. It is a parametric and distribution free Hypothesis test which verifies whether two samples are drawn from the same distribution.

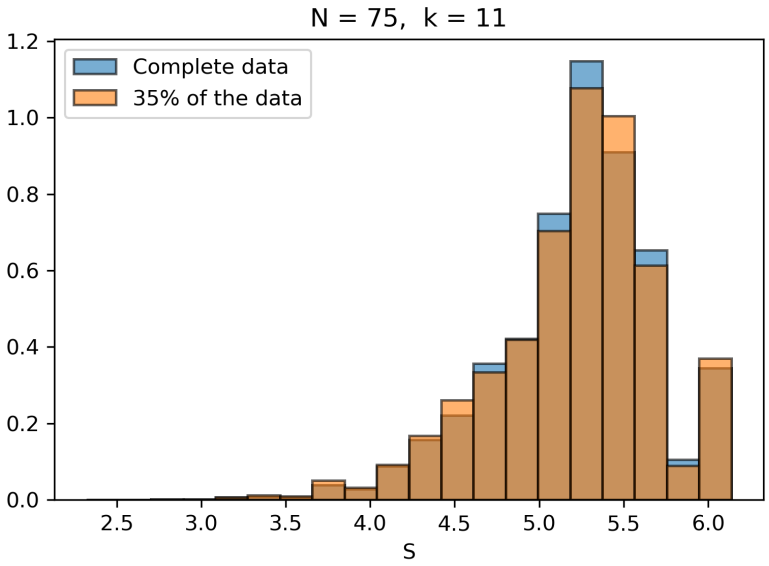


Figure 6.1: Partial data compatibility for $N = 75$, $k = 11$ using only 35% of the data

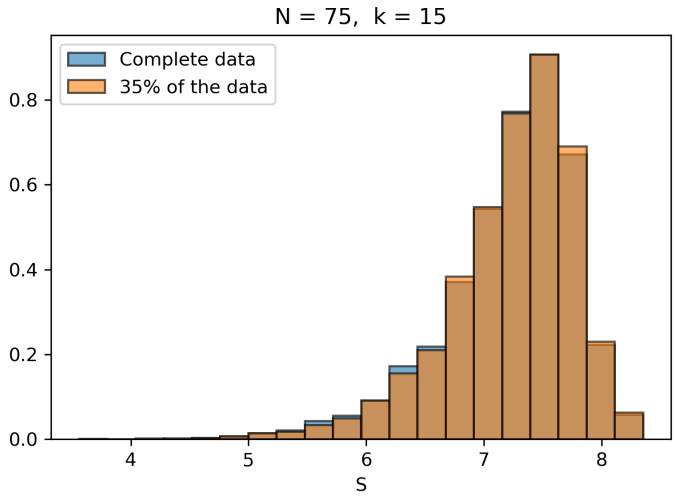


Figure 6.2: Distribution convergence for $L = 8, k = 17$ using only 35% of the data

ii. Consider the histograms and looking at the compatibility of the underlying binomial distributions

The two-sample Kolmogorov-Smirnov test for $\alpha = 5\%$ confirms the hypothesis of independently sampled data coming from the same distributions for all the computational steps.
We then test whether the data converges to the same distribution for independent simulations. We again select $N = 255$, $L = 8$, $Y = 13$, $r = 4$ and $N = 75$, $L = 7$, $r = 20$ in order to test both more peaked distributions and different system sizes.
As we can see in Figures 6.3 and 6.4, independent simulations result in compatible distributions. Compatibility of the underlying binomial distributions is verified for all computational steps.

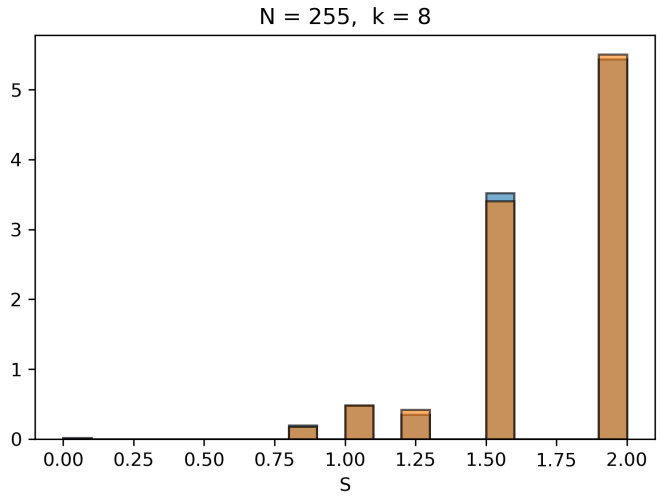


Figure 6.3: Independent simulations for $N = 255$, $L = 8$, $k = 8$. Each color represents an independent run.

Both criteria are met for entanglement entropy distributions for all computational steps, both for $N = 255$ and $N = 75$, so we can effectively use Montecarlo methods in order to simulate systems with $k+L > 19$. This allows us to look at entanglement entropy dynamics for systems with $L \geq 7$.
The memory requirements are still prohibitive for 30 qubits systems[3], so we can compute the entanglement entropy distributions for up to 27 qubits systems, or $L = 9$.

[3]Even if the memory requirements were not prohibitive, the computational time needed to perform the IQFT (once) and the spectral decomposition (for each bipartition) would allow us to only consider a bipartition subset of order 10^2 at most, out of a total bipartition subspace of order 10^8

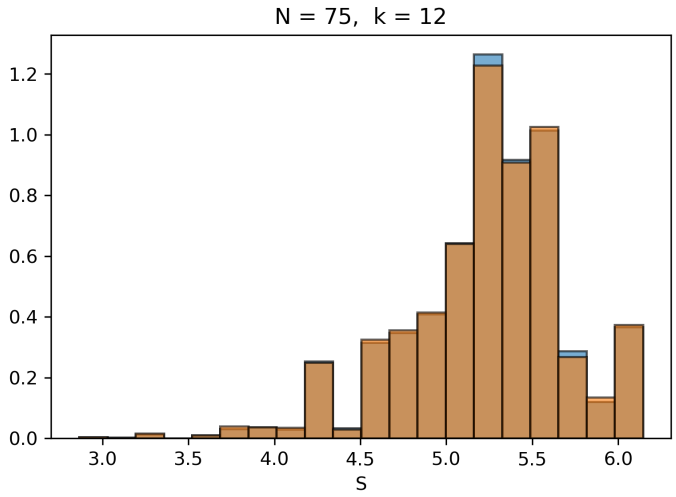


Figure 6.4: Independent simulations for $N = 75$, $L = 7, k = 12$. Each color represents an independent run.

7. Entanglement entropy mean behaviour

We've seen that the entanglement entropy between registers follows a specific pattern: for fixed parameters Y, N and consequent order r, the entanglement entropy grows for approximately $\log_2(r)$ computational steps. After that step, the entropy is almost constant: $S(k|r)) \approx \log_2(r)\ \forall k \geq \text{ceil}(\log_2(r))$ (cfr. Figure 4.1).
We therefore stated that the entanglement entropy between registers only depends on the parameter r.
We want to test whether a similar behaviour can be found for the entanglement entropy across all the possible bipartitions, or if we can link its dynamics to some property of N, Y or r.

7.1 Mean entanglement entropy across computational steps: $\mu_S(k)$

We computed the mean entanglement entropy $\mu_S(k|N, r)$ in order to observe its behaviour across Shor's algorithm. The mean is taken with respect to all the values of S for the different balanced bipartitions of a $k + L$ qubits system. We used Montecarlo methods $\forall k + L \geq 19$ and $Y = 13$ is fixed. In order to simplify the notation we will denote by $\mu_S(k)$ the mean entanglement entropy for a fixed N, Y pair.
As we see in Figure 7.1, $\mu_S(k)$ seems to follow a sigmoid behaviour for $k \in \{1, .., 2L\}$: it grows following a logarithmic-like curve up to a certain k^* and then it remains almost constant. This is very similar to what happened for the registers bipartition. The IQFT introduces a spike in the mean entanglement, which is given by the entanglement generated by the IQFT on the control register.
We will look at the behaviour of μ_S without the IQFT, as represented in Figure 7.2 We can make the following observations.

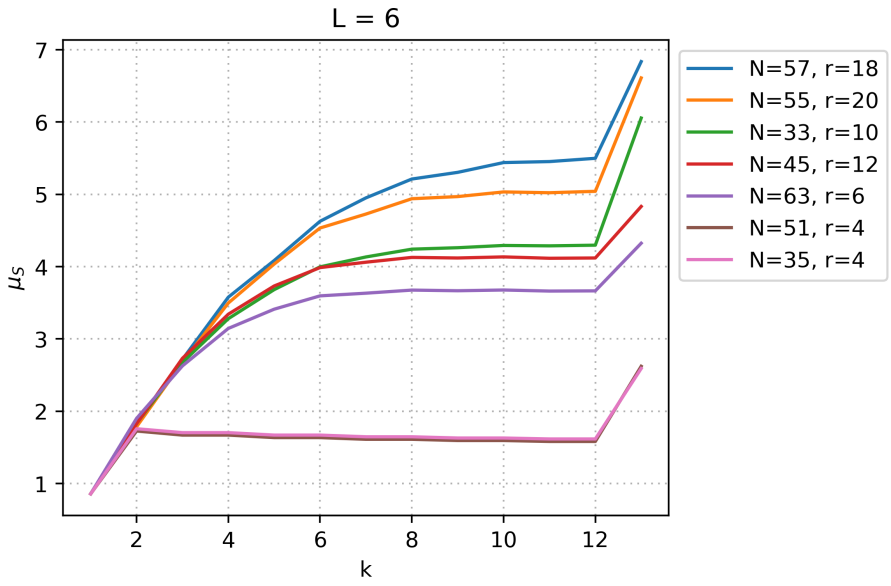


Figure 7.1: $\mu_S(k)$ for $Y = 13$ and $N \in \{33, .., 64\}$ which are suitable for Shor's algorithm, i.e. N odds which are not prime nor prime powers. IQFT included, all bipartitions have been considered

- The mean entropy can decrease with the computational step. This never happened for the entanglement dynamics between registers. A significant decrease is observed only for $r = 4$, where we find an 8% decrease of μ_S.
- μ_S is not monotone as a function of r, unlike the register entropy evolution, where $S(r) \propto \log_2(r)$.
- The number of computational steps needed to reach the horizontal asymptote is no more ceil($\log_2(r)$). As an example, for $N = 57, Y = 13, r = 18$ the mean does not converge until $k \geq 10$, despite $\log_2(18) < 5$.
- The number of computational steps needed to reach the horizontal asymptote is no more directly proportional to r: for $Y = 13$, we observe $N = 33, r = 10$ reach the maximum value of μ_S at a greater computational step than $N = 45, r = 12$.

We therefore cannot generalize the dynamics of the entanglement entropy between the registers to the arbitrary case.
We now want to test whether the mean entanglement entropy is a function of the order r only, as in the registers bipartition. In order to verify this and gain further

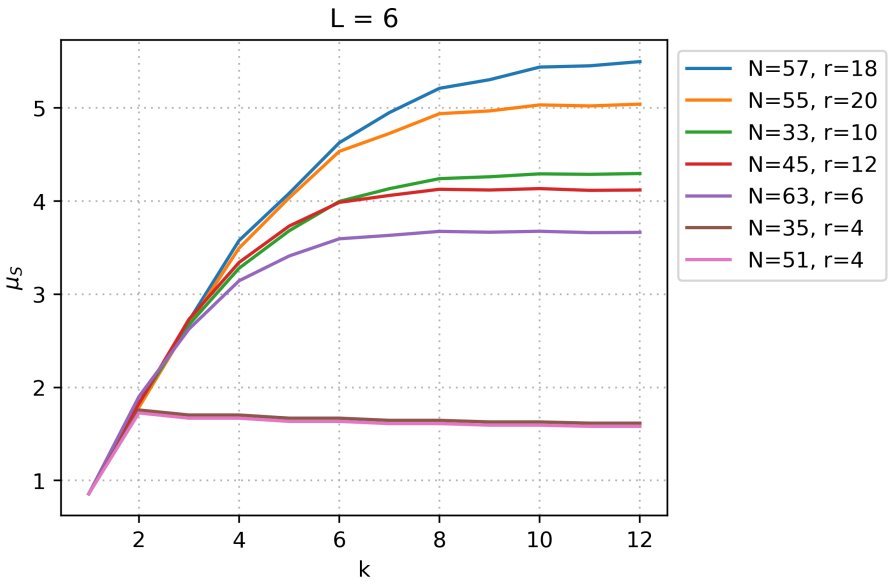


Figure 7.2: $\mu_S(k)$ for $Y = 13$ and $N \in \{33, .., 64\}$ which are suitable for Shor's algorithm, i.e. N odds which are not prime nor prime powers. IQFT not included. All bipartitions have been considered.

insights, in Figure 7.3 we show $\mu_S(k)$ for fixed $Y = 13$ and all suitable $N \in \{65, 128\}$ $(L = 7)$.
For $L = 7$, the peculiar behaviour of N, Y pairs with order $r = 4$ is repeated. μ_S seems not to be proportional to either r or N. Since we notice a peculiar trend for pairs with order $r = 4$, we now look at $Y = 13, N \in \{93, 99, 111, 95\}$ in order to observe the evolution of $\mu_S(k)$ for some same-ordered pairs with $r \neq 4$.
Since the data in Figures 7.4 - 7.5 shows that same-ordered pairs have equal or similar behaviour, we can conclude that the order r does not completely characterize the behaviour of $\mu_S(k)$, but it plays a strong role in it.

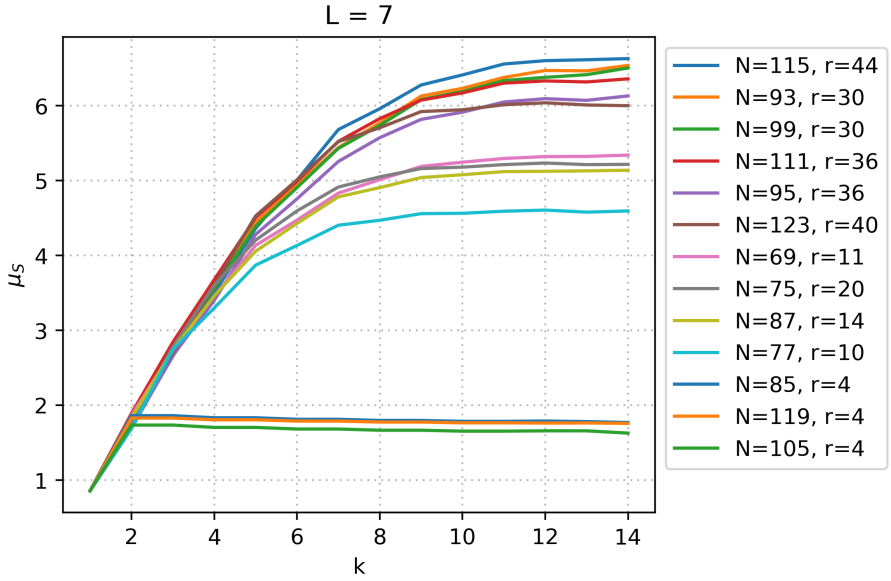


Figure 7.3: $\mu_S(k)$ for $Y = 13$ and $N \in \{65, .., 128\}$ which are suitable for Shor's algorithm, i.e. N odds which are not prime nor prime powers. IQFT not included. Montecarlo methods on the distribution space have been applied

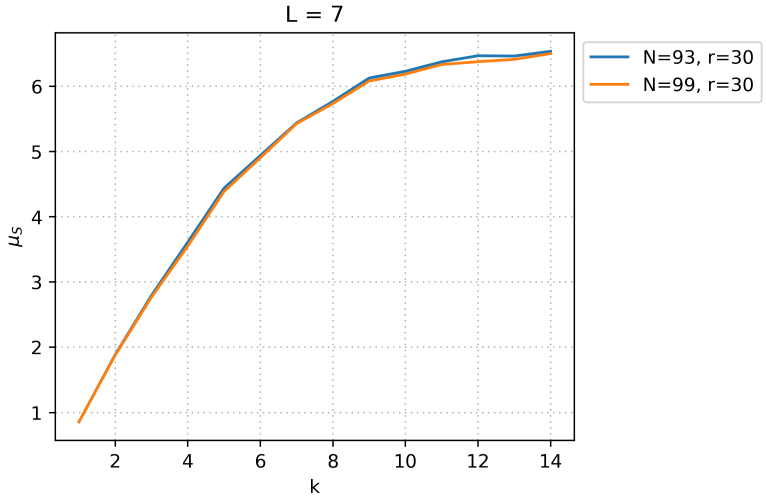


Figure 7.4: $\mu_S(k)$ for same ordered pairs $(93, 13)$, $(99, 13)$ with $r = 30$

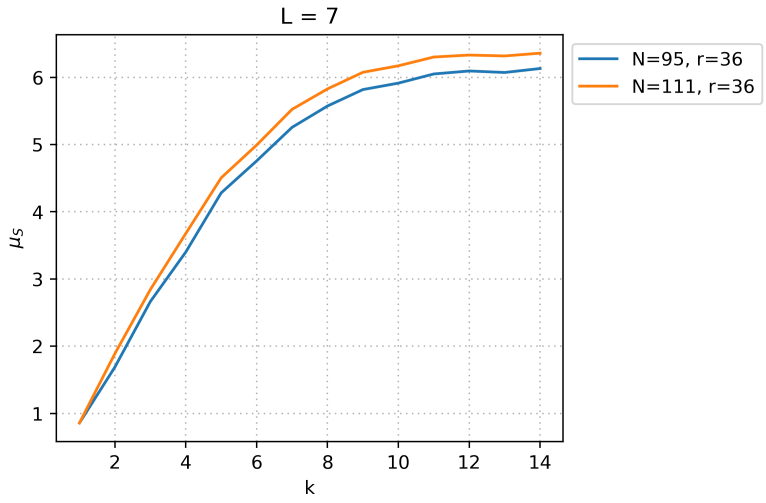


Figure 7.5: $\mu_S(k)$ for same ordered pairs $(95, 13)$, $(111, 13)$ with $r = 36$

7.1.1 Entanglement entropy for $r = 2^n$

We noticed that the mean entanglement entropy dynamics for $r = 4$ has a very peculiar behaviour: it saturates at $k = 2$ to a maximum value smaller than 2, then slightly decreases as a linear function of the computational step.
We try to see if this behaviour is repeated for some other (Y, N) pair such that $r = 2^n$ for some $n \in \mathbb{N}$. The results are shown in Figure 7.6.
As we can see, the curves behaviour is similar: if we normalize each curve into $\mu_S(N) \in [0, 1]$ and properly scale the k axis, as we did in Figure 7.7, the curves all decrease linearly after reaching a maximum, with a linear coefficient which is slightly variable with N.
We can conclude that for orders which are powers of two, the decay of the mean entanglement entropy is almost constant with respect to r.
If we compare all these results to the one obtained for the registers case, we can summarize the entanglement entropy dynamics as such:

- For all $k \in \{1, .., \text{ceil}(\log_2(r(Y, N)))\}$, the entanglement grows almost linearly. This is similar to the dynamics of registers bipartition. Since the mean entropy with respect to all possible bipartitions is roughly equal to the one for the registers case, we suppose the entanglement entropy between control and target register is almost evenly distributed among the target register qubits.

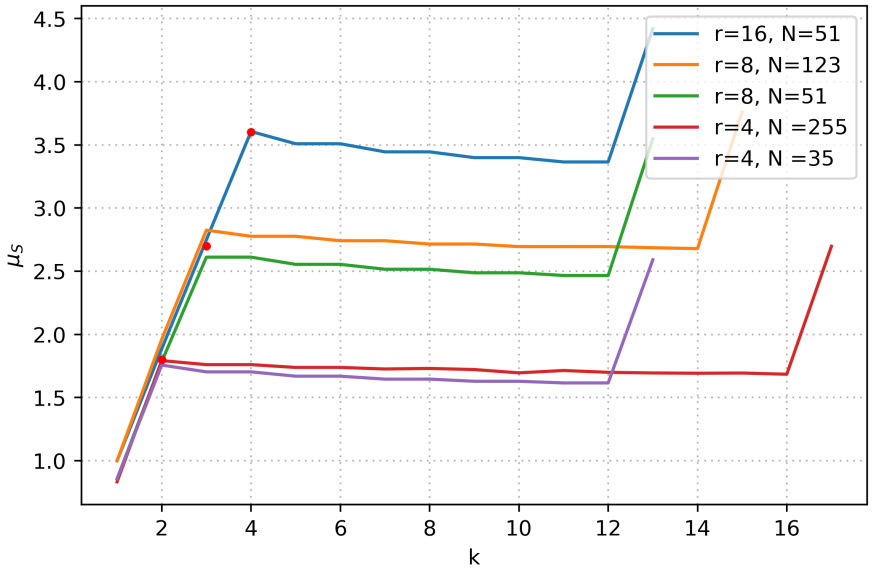


Figure 7.6: The curves show $\mu_S(k)$ for (Y, N) pairs such that r is a power of two. The red dots represents $9r/10$. All bipartitions have been considered for $N \leq 51$

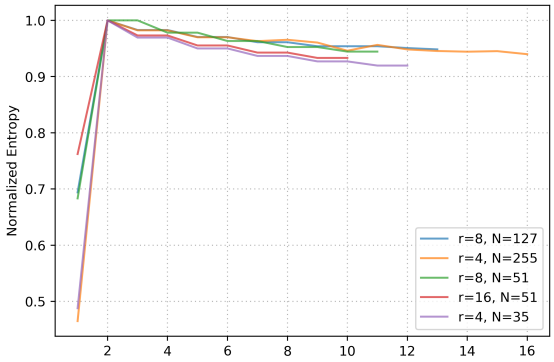


Figure 7.7: Normalized and scaled version of Figure 7.6: decay of $\mu_S(N, k)$ for $k > \log_2(r)$ is almost constant with respect to of r, N

- If $r(Y, N)$ is a power of 2, the mean entanglement entropy reaches its maximum value at

$$\mu_S(N, k = \log_2(r)) \approx \frac{9 \log_2(r)}{10} + C(N)$$

.

The maximum value appears to depend mostly on r, with little oscillations $C(N) \ll \log_2(r)$. In Figure 7.6 we plotted $9\log_2(r)/10$ as red dots: the maximum value is often very close to our approximation and the oscillations for $r = 8$ are symmetrical. For $k > \log_2(r)$, $\mu_S(N,k)$ decays with an almost constant rate, as we show in Figure 7.7.

- If $r(Y,N)$ is not a power of 2, the mean entanglement entropy never present a significant decrease: entanglement is generated in the target register and the curve $\mu_S(N,k)$ can be approximated using a sigmoid function:

$$\mu_S(N,k,r) = \frac{A}{1+e^{-B(k-C)}} \tag{7.1}$$

By considering a normalized version of the graphs in Figures 7.4 - 7.5 and the estimated sigmoid parameters in Table 7.1, we can conclude

$$\mu_S(N,k,r) \approx \frac{A(N,r)}{1+e^{-B(r,L)\,(k-C(r,L))}}$$

For fixed L, the rate of growth B and the scale C only depends on the order r, the maximum value A of $\mu_S(k)$ instead depends on both r and N.

N	L	r	A	B	C
183	8	3	2.59±0.01	0.98±0.03	1.67±0.03
63	6	6	3.65±0.02	0.98±0.04	2.04±0.04
33	6	10	4.27±0.03	0.80±0.03	2.46±0.05
77	7	10	4.57±0.03	0.77±0.03	2.68±0.05
231	8	10	4.93±0.03	0.73±0.03	2.75±0.06
69	7	11	5.29±0.04	0.64±0.03	3.05±0.07
45	6	12	4.11±0.02	0.91±0.03	2.31±0.03
153	8	12	4.40±0.01	0.87±0.02	2.48±0.03
159	8	13	5.89±0.04	0.61±0.02	3.36±0.07
87	7	14	5.10±0.03	0.68±0.02	2.94±0.06
147	8	14	5.47±0.04	0.67±0.03	3.04±0.06
203	8	14	5.56±0.03	0.66±0.02	3.13±0.06
57	6	18	5.44±0.06	0.64±0.03	3.14±0.08
133	8	18	5.78±0.05	0.60±0.03	3.33±0.08
171	8	18	5.77±0.04	0.59±0.02	3.36±0.08
189	8	18	5.97±0.05	0.61±0.03	3.30±0.08

N	L	r	A	B	C
55	6	20	5.02±0.03	0.72±0.02	2.90±0.05
75	7	20	5.20±0.02	0.73±0.02	2.93±0.04
165	8	20	5.45±0.02	0.70±0.02	3.13±0.04
175	8	20	5.61±0.03	0.68±0.02	3.11±0.05
187	8	20	5.53±0.02	0.68±0.02	3.19±0.04
129	8	21	6.55±0.06	0.53±0.02	3.83±0.09
161	8	22	6.27±0.04	0.61±0.02	3.56±0.07
145	8	28	6.01±0.04	0.60±0.02	3.49±0.06
245	8	28	6.30±0.03	0.62±0.02	3.51±0.05
93	7	30	6.46±0.07	0.54±0.02	3.68±0.09
99	7	30	6.42±0.06	0.54±0.02	3.69±0.09
217	8	30	6.87±0.06	0.52±0.02	3.93±0.09
207	8	33	7.39±0.11	0.44±0.03	4.28±0.14
95	7	36	6.08±0.04	0.58±0.02	3.62±0.06
111	7	36	6.32±0.05	0.59±0.02	3.52±0.07
135	8	36	6.30±0.05	0.57±0.02	3.59±0.08
185	8	36	6.57±0.05	0.56±0.02	3.67±0.08
237	8	39	7.95±0.09	0.46±0.02	4.50±0.11
123	7	40	6.02±0.03	0.66±0.01	3.38±0.04
205	8	40	6.11±0.03	0.63±0.02	3.38±0.05
115	7	44	6.62±0.04	0.56±0.01	3.87±0.05
141	8	46	7.44±0.08	0.48±0.02	4.24±0.10
177	8	58	7.91±0.09	0.46±0.02	4.49±0.11
155	8	60	7.16±0.07	0.49±0.02	4.15±0.09
225	8	60	7.39±0.07	0.50±0.02	4.18±0.10
201	8	66	7.86±0.12	0.44±0.02	4.51±0.14
213	8	70	8.29±0.10	0.44±0.02	4.75±0.11
219	8	72	7.20±0.05	0.53±0.02	4.04±0.07
249	8	82	8.48±0.11	0.44±0.02	4.80±0.12
215	8	84	7.94±0.08	0.46±0.02	4.53±0.10
209	8	90	8.58±0.11	0.41±0.02	5.04±0.12
235	8	92	8.18±0.10	0.45±0.02	4.59±0.12
253	8	110	8.86±0.13	0.41±0.02	5.21±0.14

Table 7.1: Table of fitted values for parameters in Equation 7.1. $Y = 13$ is fixed

7.1.2 Entanglement after modular exponentiation

In previous sections we did not take in account the post-IQFT entanglement.
We will now explain why entanglement after the modular exponentiation cannot be considered interesting for measuring algorithmic complexity. In order to do this, we need to consider an equivalent formulation of Shor's algorithm.
The modified algorithm is a consequence of the principle of implicit measurement, as in Reference [7]:

> **Principle of implicit measurement**
> Without loss of generality, any unterminated wire at the end of a quantum circuit may be assumed to end with a measurement

For Shor's algorithm, this means that we can measure the target register after the modular exponentiation without affecting the algorithm result.
More generally, measuring the state of either subsystem (here, registers) destroys the entanglement between the subsystems. Indeed, if we consider a bipartite system in state $|\psi\rangle = \sum_x |x\rangle |f(x)\rangle$ and we measure the second subsystem, let denote by y the state associated with the measurement result, the whole state collapses to

$$\left(\sum_{x:y=f(x)} |x\rangle \right) \otimes |y\rangle \tag{7.2}$$

which is a separable state, therefore the entanglement entropy between registers is 0.
For Shor's entanglement, if we define $f(x) = Y^x \bmod N$,the same structure applies.
Similarly to what happens in measurement-based quantum computation, here the measurement operation consumes the entanglement between the registers in order to transform the global state. The collapse of the target wave function effectively forces the system to immediately exploit the quantum mutual information between registers.
An intuitive explanation for the implicit measurement principle in Shor's algorithm can be obtained by considering a quantum information framework: let us have the registers of Shor's algorithm in state $|\psi\rangle_{2L}$. At this point the control register is given to Alice and the target register is given to Bob. Bob can decide whether and when to

measure the target register. Alice can decide when to apply the IQFT and measure the control register.
Suppose Bob travels five light-years away from Alice: if Bob's measurement could affect in any way the result of Alice's operations, we would have an information exchange between the two parties which could be faster than light. Since faster than light communication is not possible, the measurement of the target register cannot affect the result of Shor's algorithm.
Applying a measurement before the IQFT highlights the purpose of entanglement between registers in the computation. A similar analysis cannot be easily carried out for the other entanglement source, i.e. entanglement within the target register.
As we show in Appendix A, although we are able to compute the entanglement for a subset of the target register, this entanglement is always referred to the whole $3L$ qubits. Defining an entanglement within the target register would require using a multipartite measure of entanglement, which, as we suggested in Section 2.3, is a rather complex task.
From the data in Figure 7.1, we observe that the IQFT causes a spike in the mean entanglement. This is to be expected, since the IQFT operation only affects the control register and bring it from a product state to a generic (possibly) entangled state. We will now show that the there exists an implementation of Shor's algorithm in which the entanglement generated during the IQFT has been proven to be negligible.
We recall the properties of the modular exponentiation with respect to the order:

$$f(x) = Y^x \bmod N = t = f(x + lr) \quad \forall l \in \mathbb{N}^+$$

Thus if we consider the modified Shor's algorithm where the target register has been measured and collapsed into the state $|f(x_0)\rangle$, we can rewrite the control state in Equation 7.2 as

$$\sum_{l=0}^{v} |x_0 + l \cdot r\rangle$$

where $v = \text{floor}((2^{2L} - 1)/r)$ and $x_0 \in \{1, ..., r - 1\}$ such that $Y^{x_0} \bmod N = f(x_0)$.
States of this form are called periodic states and it has been shown in Reference [26] that the IQFT operation on periodic states does not significantly modify the entanglement of incoming states. Furthermore, it is proved in Reference [27] that periodic states generate multipartite entanglement in unbounded many qubits.
Since there exists an implementation of the algorithm for which the entropy does not change during the IQFT operation, we decided not to model the entanglement

entropy after the modular exponentiation. Despite this, we still calculated the entanglement entropy for the $k = 2L + 1$ step in order to both verify the algorithm results and test whether the entropy invariance under IQFT is present in all implementation of Shor's algorithm.

7.2 Convex hull and averaged mean entropy

We now repeat the study of the convex hull we made for the registers bipartition in Section 4.2.2, in order to confront the results with the arbitrary case.
Computing the entropy for all Y coprimes of N is very expensive, so we can only do it for $N \leq 129$. This limitation is due to both the increasing number of coprimes for growing N and the increased dimensionality of the problem for $L = 8$.
We show a scatter plot of all values of $\mu_S(N, Y)$ for all values of N and Y in Figure 7.8.

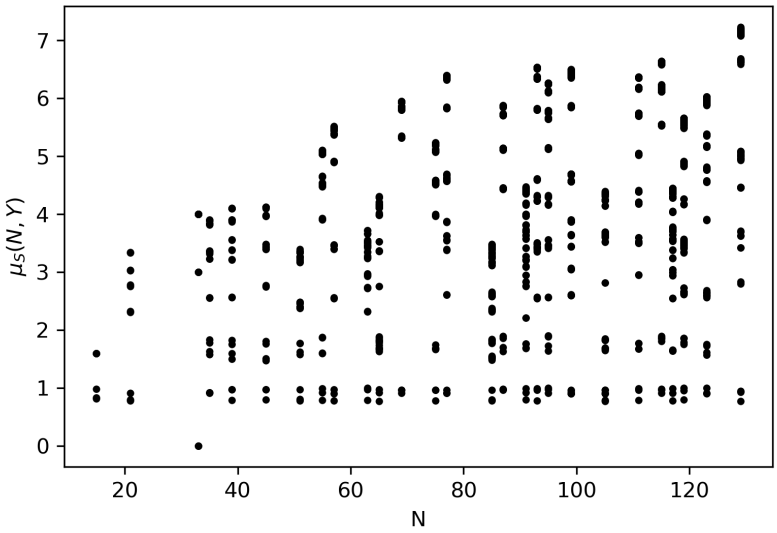


Figure 7.8: Scatterplot of $\mu_S(N, Y, k = 2L)$, $N \leq 129$

The limited amount of data makes it hard to spot a trend. We try looking if the logarithmic behaviour of the registers' entropy can be extended for the arbitrary case: we define the convex hull

$$\hat{\mu}_S(N) = \max_Y \mu_S(N, Y)$$

The data for $\hat{\mu}_S(N)$ is shown in Figure 7.9. Like in the registers bipartition results, the convex hull does not present a regular behaviour.
In Figure 7.10, we try to identify the extreme points and fit a logarithmic curve.

This leads to an approximate convex hull

$$\hat{S} \approx 1.56 \log_2(N) - 3.62$$

but the limited data makes this operation not reliable.

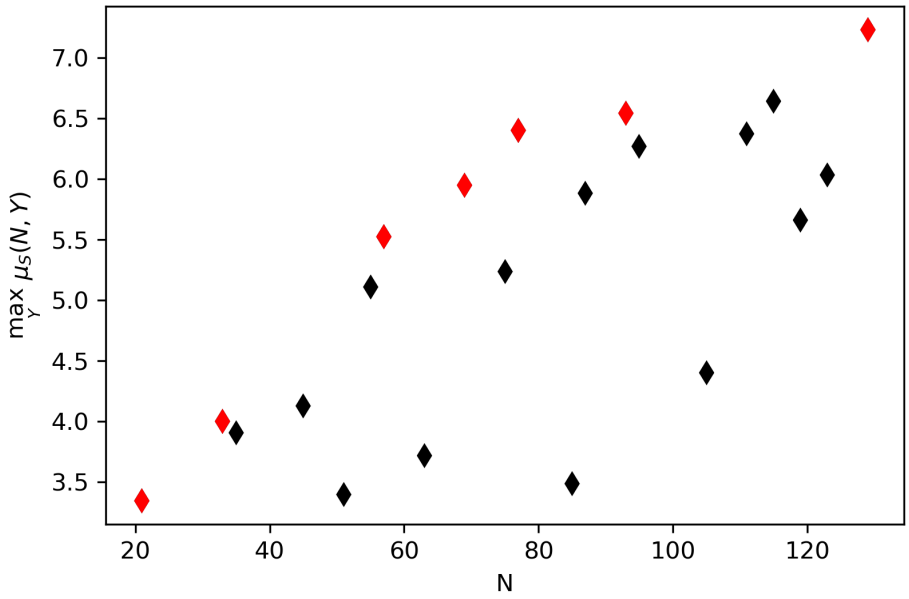


Figure 7.9: Convex hull $\hat{\mu}_S(N)$ for N s.t. $L \geq 7$. The red data represents the considered extreme points

As we did for the registers case, we first averaged $\mu_S(N, Y)$ over the coprimes Y for each N, then we divided the domain $\{15, ..., 129\}$ into subdomains N_i of length *step*. We obtain the values

$$\mu_S(N_i) = \underset{N \in N_i}{\text{avg}} \underset{Y:GCD(Y,N)==1}{\text{avg}} \mu_S(Y, N)$$

We test whether this values follows a logarithmic curve as a function of N. In Figures 7.11 to 7.13, we show in semi-logarithmic scale the values of $\mu_S(N_i)$ for different values of the *step* variable and we compare them to the logarithmic fit obtained for the register case.
The parameters of the linear mean are again weakly dependent on the *step* parameter: the linear fit gives

$$E[\mu_S(N, step)] \approx 1.18 \log_2(N) - 3.2$$

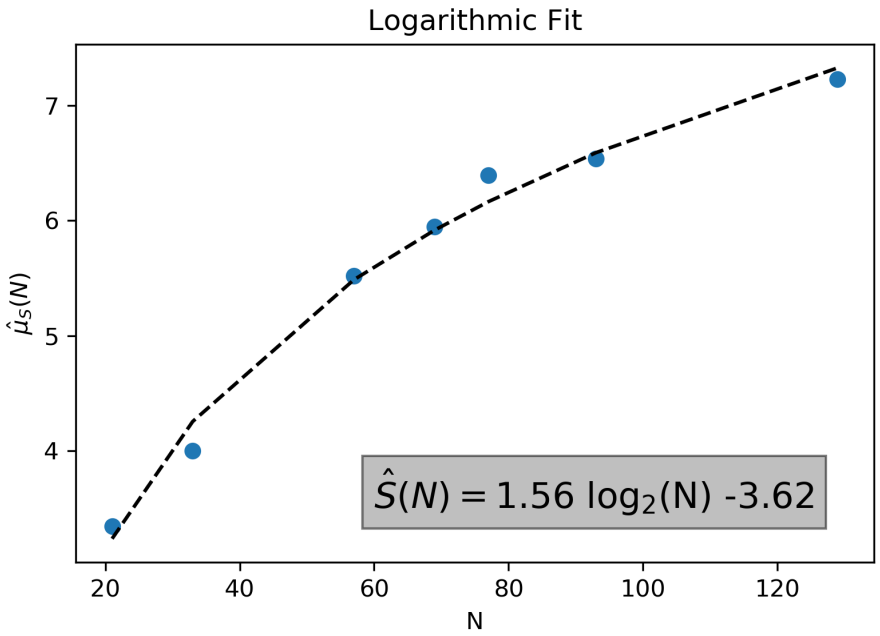


Figure 7.10: Logarithmic fit of convex hull's selected points

In Figures 7.11 to 7.13 we confronted the arbitrary bipartition averaged entropy with the registers bipartition results, contained in Figures 4.6 to 4.9.
We point out that the entanglement entropy is an extensive measure: given a system of m qubits in a pure state, the entanglement entropy of an n qubits subsystem is bounded by

$$S \leq \min(n, m - n)$$

At the end of the modular exponentiation, let L be the number fo qubits in the target register, we have $S_{reg} \leq L$ for the entanglement entropy between registers and $S_{arb} \leq \text{floor}(3L/2)$ for the arbitrary case. It is therefore consistent that in our data the arbitrary bipartitions, formed by more qubits, have a greater averaged entanglement entropy.
We normalized the entropy measures by dividing for the maximal entropy and confront the two results in Figures 7.14 to 7.16.

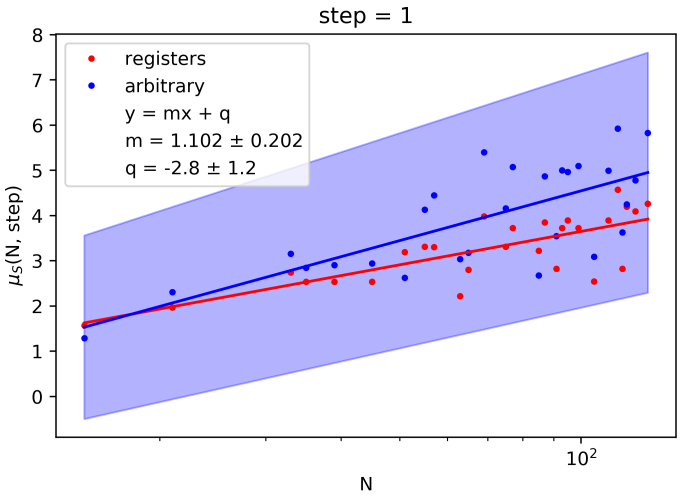


Figure 7.11: In blue we show $\mu_S(N_i)$ for arbitrary bipartitions with estimated linear parameters, in red we plot this quantity for the registers bipartition. $step = 1$. The blue area contains data points within one standard deviation from the mean (blue linear fit).

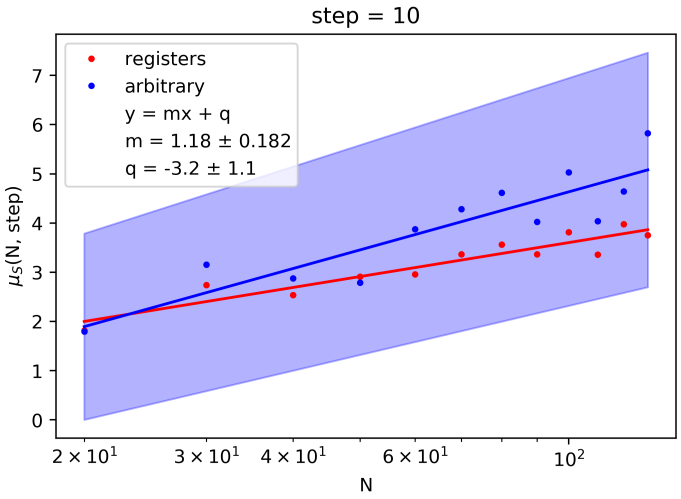


Figure 7.12: In blue we show $\mu_S(N_i)$ for arbitrary bipartitions with estimated linear parameters, in red we plot this quantity for the registers bipartition. $step = 10$. The blue area contains data points within one standard deviation from the mean (blue linear fit).

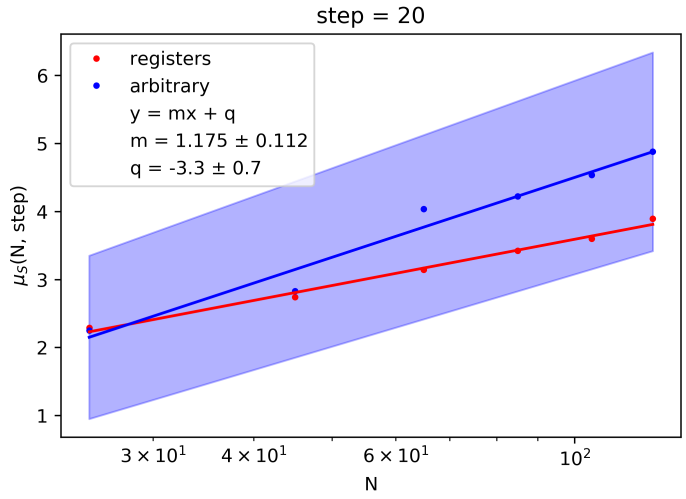


Figure 7.13: In blue we show $\mu_S(N_i)$ for arbitrary bipartitions with estimated linear parameters, in red we plot this quantity for the registers bipartition. $step = 20$. The blue area contains data points within one standard deviation from the mean (blue linear fit).

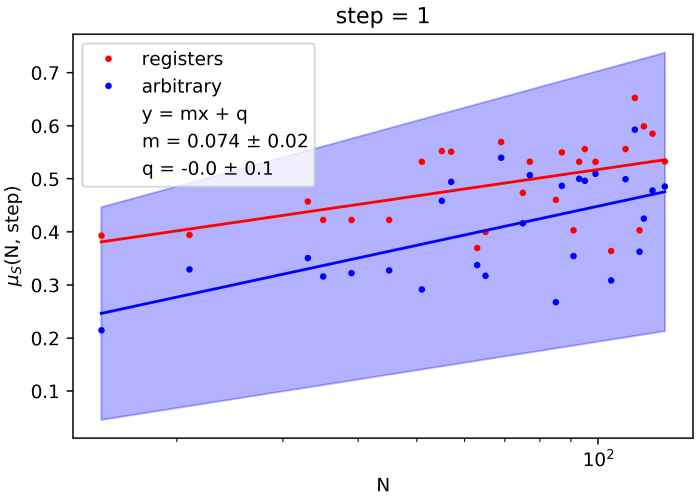


Figure 7.14: In blue we show the normalized $\mu_S(N_i)$ for arbitrary bipartitions with estimated linear parameters, in red we plot this quantity for the registers bipartition. $step = 1$. The blue area contains data points within one standard deviation from the mean (blue linear fit).

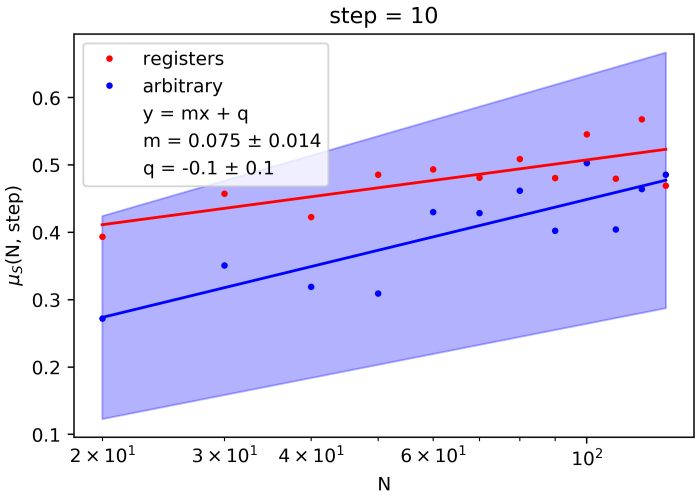


Figure 7.15: In blue we show the normalized $\mu_S(N_i)$ for arbitrary bipartitions with estimated linear parameters, in red we plot this quantity for the registers bipartition. $step = 10$. The blue area contains data points within one standard deviation from the mean (blue linear fit).

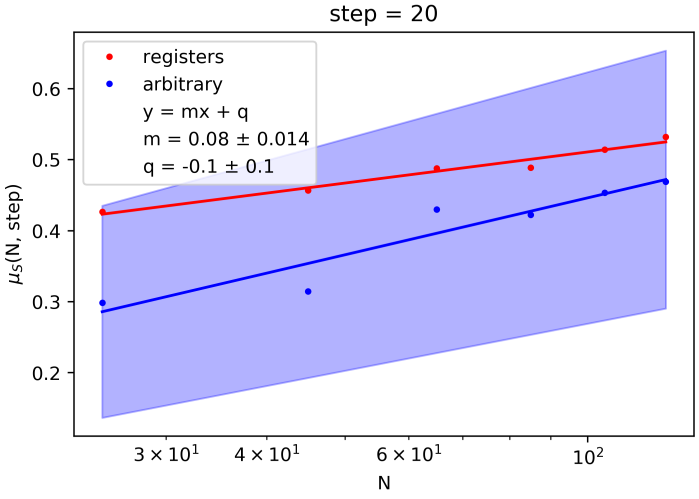


Figure 7.16: In blue we show the normalized $\mu_S(N_i)$ for arbitrary bipartitions with estimated linear parameters, in red we plot this quantity for the registers bipartition. $step = 20$. The blue area contains data points within one standard deviation from the mean (blue linear fit).

We underline that for all data points $\forall\, step\, \mu_S^{arb}(N, step) < \mu_S^{reg}(N, step)$. This means that, on average, the entanglement entropy for the registers case can be an upper bound for the arbitrary case for $N \leq 129$. An estimate for the arbitrary normalized entanglement entropy is reported in Figures 7.14 to 7.16, but the variance of the data makes this fit very unreliable.
We also note that the arbitrary averaged entropy grows more rapidly than the averaged registers'.
Taking in account that there can be no entanglement within the control register at the computational step $k = 2L$, this behaviour suggests that in the general case there is entanglement in the target register, whose amount grows (averaging on Y and over subdomains N_i) with N. This is to be expected, since most of the composite Hilbert space is made of entangled states and the number of qubits in target register grows with N.

8. Entanglement entropy distributions

Until now we only looked at the mean of entanglement entropy with respect to all the possible balanced bipartitions. We now consider the whole distribution of entanglement entropy for fixed N, Y, r across all bipartitions. In this Chapter we investigate the relationship between the order and the distribution.

8.1 Entanglement entropy dynamics in bipartitions space: role of the order

Using GPU accelerated eigenvalues methods for Hermitian matrices, we got the following entropy distributions. We fixed $Y = 13$ and let the parameter N vary.

8.1.1 L = 4

For $L = 4$ we are able to evaluate the entanglement entropy for all the balanced bipartitions. The following normalized results are obtained for the only suitable pair of numbers $N = 15$, $Y = 13$, which has order $r = 4$.

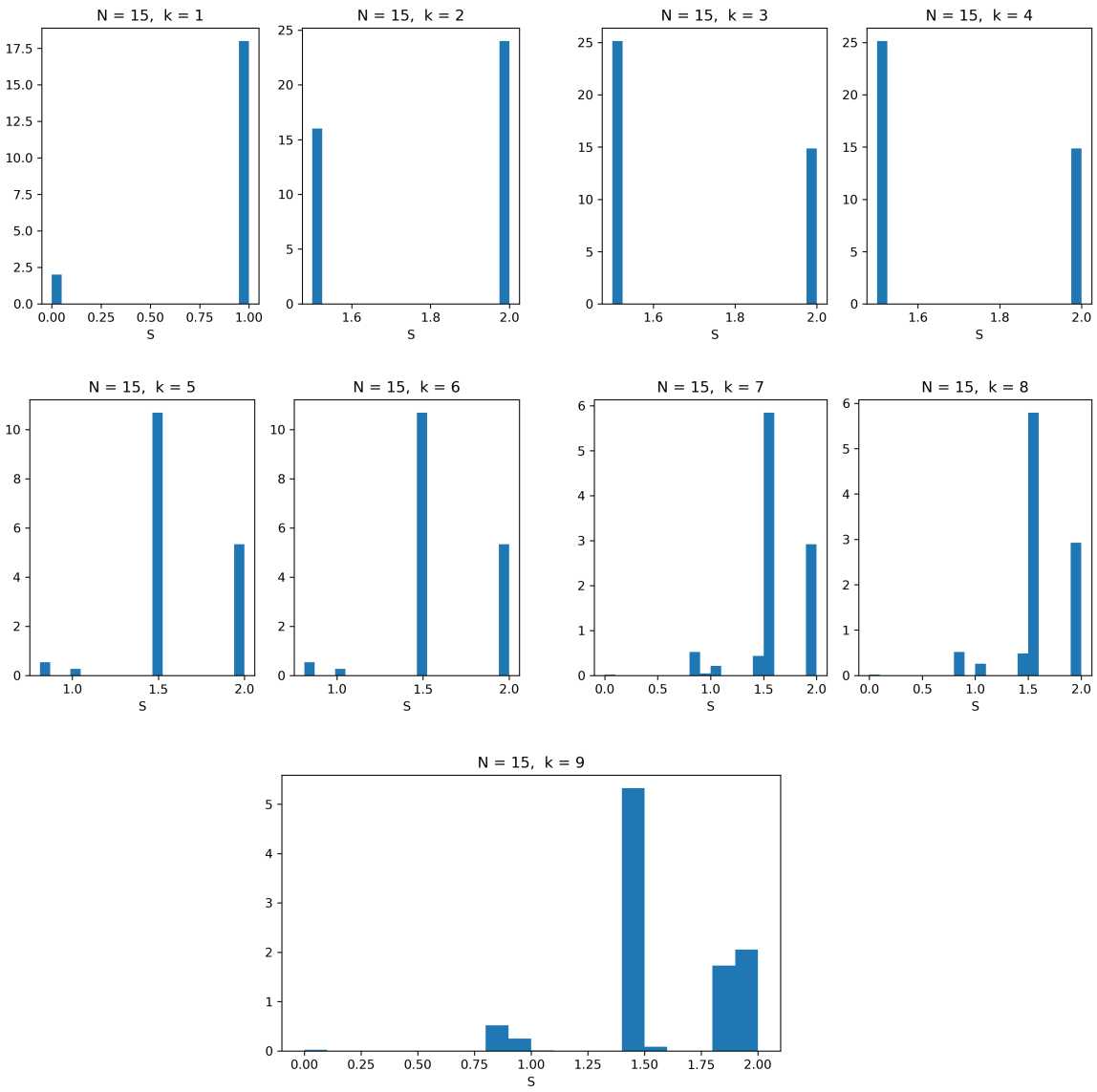


Figure 8.1: $N = 15$, $Y = 13$, $r = 4$

8.1.2 Entanglement entropy dynamics for fixed order

For $L = 5$ we are able to consider all the bipartitions of 3L qubits without using Montecarlo methods.

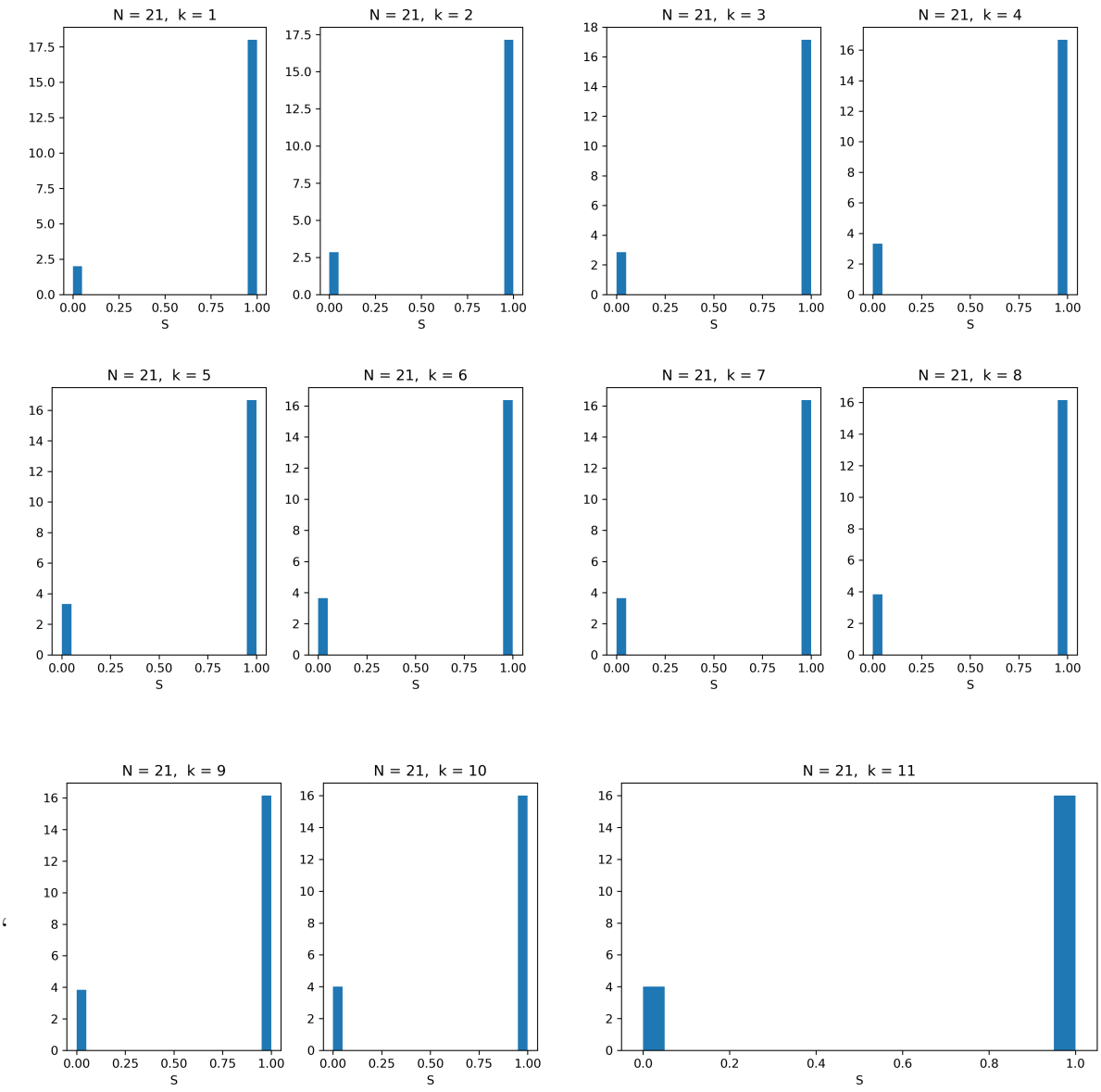
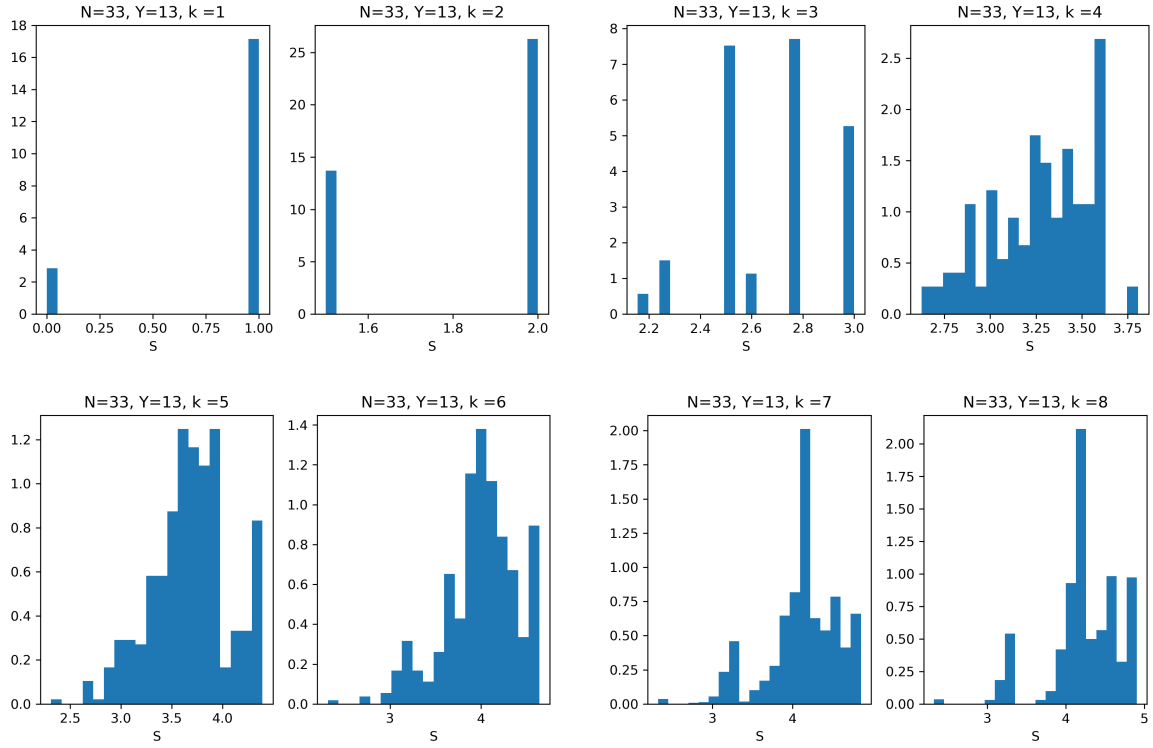


Figure 8.2: $N = 21$, $Y = 13$, $r = 2$

We can see that for this specific pair (N, Y)=(21,13) the entanglement entropy change very little with the computational step and is highly concentrated around specific values. This implies that only a few qubits are actually entangled.
Since the values of S found are either 0 or 1, depending on the chosen bipartition,

we can suppose that for $N = 21$ there are only two maximally entangled qubits in the system: the entanglement entropy is 1 when only one entangled qubit is traced away, while $S = 0$ when the two qubits are both or neither traced away.
Since $\log_2(r) = 1$ from Section 4.2 we know that there's only one control qubit maximally entangled with the target register, we can conclude that at the end of modular exponentiation there's only a single qubit in the control register maximally entangled with a single qubit in the target register.
Instead if we look at $N = 33$, which has order $r = 10$, we can see in Figure 8.3 that the (33,13) pair generate an entanglement entropy distribution which is far less localized. This tells us that for this pair of inputs, each of the qubits is entangled with the majority of the remaining ones.
As we can see in Figures 8.2 and 8.3, changing N and therefore r drastically modifies the entanglement entropy shape.



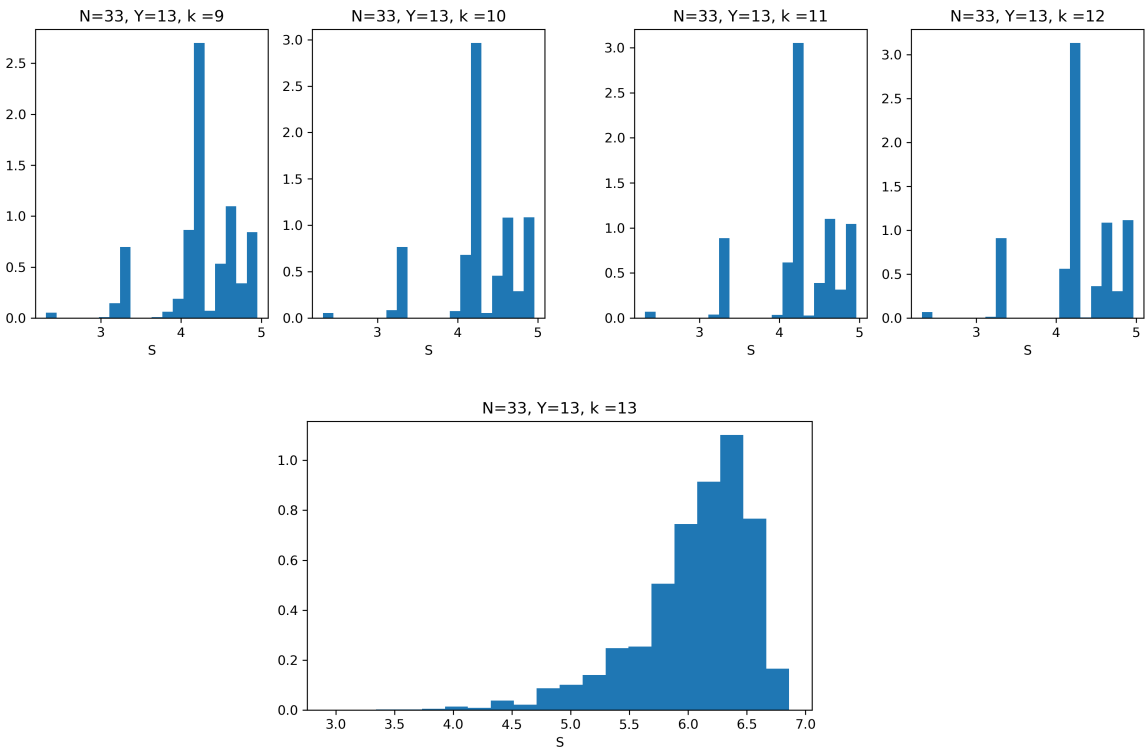


Figure 8.3: Entanglement entropy distribution for $N = 33$, $Y = 13$, r=10

In order to understand whether the order r can completely identify the shape of the entanglement entropy distribution for generic bipartitions, we considered the (165, 13), (175,13) and (187,13) pairs, which have all order $r = 20$ and have same system dimension $L = 8$.
As we observed in Table 7.1, the mean of the distribution does not depends on r only, with oscillations which depend on the specific value of N, but in Figure 8.4 we show that even basic distribution properties cannot be uniquely defined by r (and Y, here considered fixed): for computational step $k = 8$ we observe a bi-modal behaviour for $N = 175$ which is not present for $N = 165$. The distributions shape vary even more for for 24 qubits and $r = 18$ by considering $(131, 13)$, $(171, 13)$, $(189, 13)$.
Since we observed that the entanglement entropy mean behaviour strongly depends on r being a power of 2, in order to test whether r completely identify the entanglement distribution for fixed L, Y and for r of this form, in Figure 8.5 we considered $N = 35$ and $N = 51$, which have both order $r = 4$ and same number of qubits. Additionally, they have the same order as the (15, 13) pair in Figure 8.1.

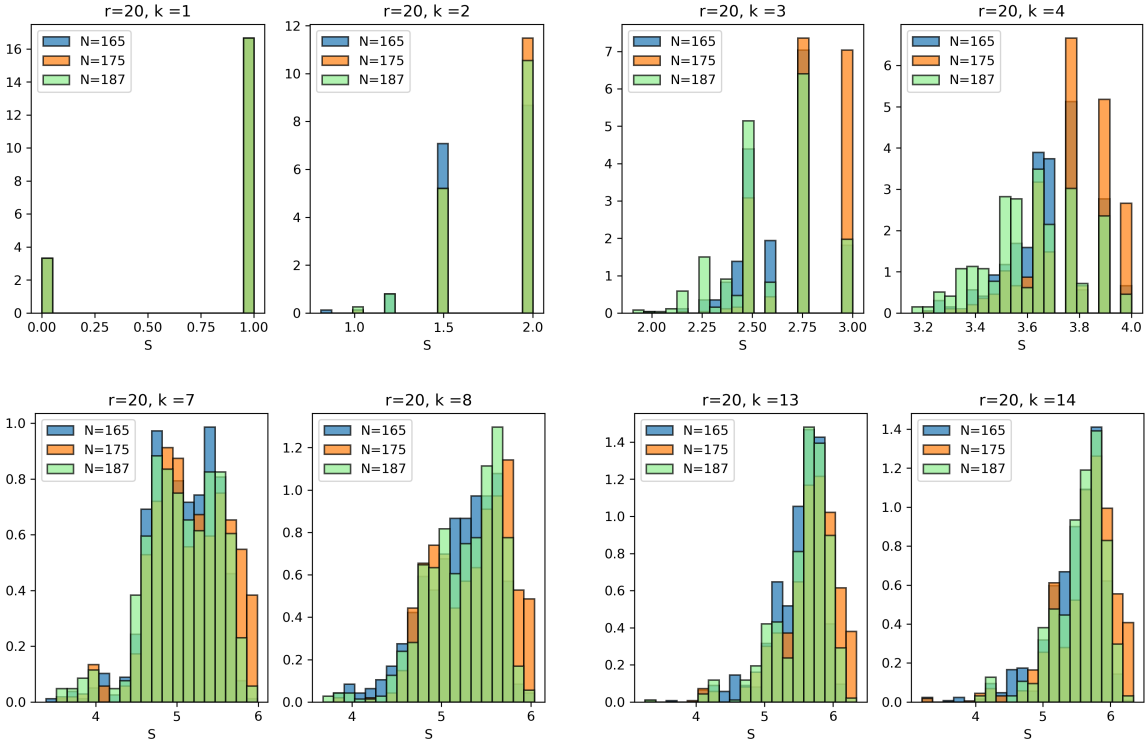


Figure 8.4: Entanglement entropy distribution for $Y = 13$, $N = 165$, $N = 175$, $N = 187$, $r = 20$

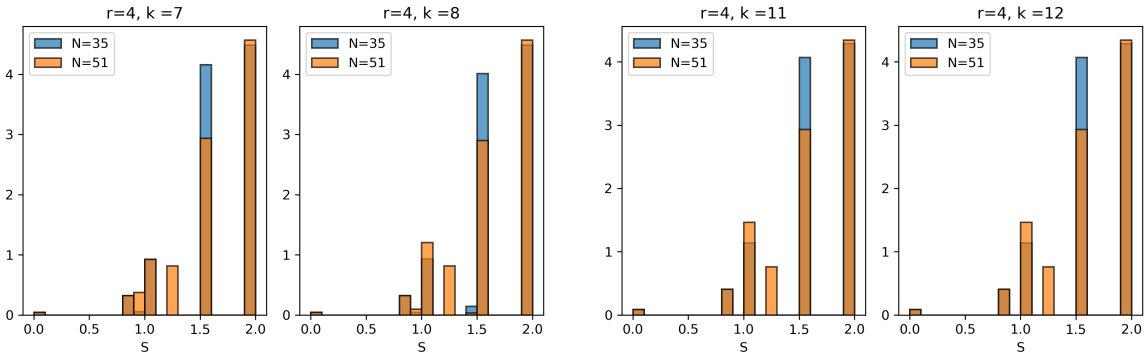


Figure 8.5: Entanglement entropy distribution for $Y = 13$, $N = 35$, $N = 51$, $r = 4$

As we can see in Figure 8.5, the entanglement dynamics can be significantly different even for (N, Y) pairs with same order $r = 2^m$ for some $m \in \mathbb{N}$.
Comparing Figure 8.5 and Figure 8.1 (same order, but different number of total qubits) seems to yield no obvious similarities between the distributions: for distinct values of L, the range of variability for S varies due to the upper bound in Equation 2.12, but even matching the scale of the distributions, the evolution of entanglement entropy for pairs with same order and different number of qubits shows no correspondence.
We can conclude that **the order does not uniquely characterize the first moments of the entanglement entropy distributions during the evolution Shor's algorithm**, even for fixed Y. Despite this, we would like to understand its role, if any, in the dynamics of S.

8.1.3 Order and multipartite entanglement

From all the previous figures, we notice that distributions with order $r \in \{2, 4\}$ are strongly peaked, while the distributions in Figure 8.3 are less localized for $r = 10$. Increasing the number of qubits to 21 ($L = 7$) and 24 ($L = 8$), we verified the pairs (105,13) (119,13) and (255,13), with same order $r = 4$, are all characterized by sharply peaked distributions.
We therefore formulate an hypothesis: the value of r is related to the number of outcomes for the random variable S, i.e. the size of the domain in which S can take values. Even by the results obtained in Section 4.2 support our hypothesis: only the first $c^* = \text{ceil}(\log_2(r))$ control qubits are entangled with the target register, therefore the number of possible outcomes of bipartite entanglement entropy among $c^* + L$ qubits grows with r for fixed L. This is due to the fact that the more qubits are entangled, the more complex the entanglement structure can develop.
This is also equivalent to stating that the entanglement becomes more and more multipartite for high values of the order.
From now on we allow ourselves to expand the domain of inputs N, since the quantum part of Shor's algorithm, i.e. the order finding subroutine, is well defined even for N even or prime. We decided to do this in order to have access to a wider range of numbers N and therefore orders r.
We confront the pairs (22,13) and (23,13), which have order $r_{22} = 10$ and $r_{23} = 11$ respectively (the Y parameter, fixed, is implicit in this notation), in order to see if they're compatible with the hypothesis of less peaked distributions for pairs associated with an higher value of r.
Both distributions in Figure 8.6 are almost continuous, which further strengthen our

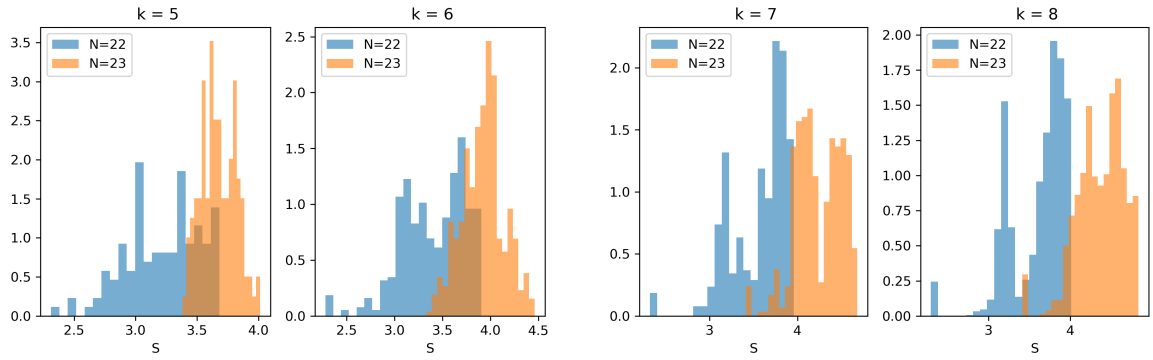


Figure 8.6: $Y = 13$, $N = 22$, $N = 23$, $r_{22} = 10$, $r_{23} = 11$

hypothesis. The same has been observed even for the pair (33,13), which corresponds to the same order $r_{33} = 10$, but in a system of 18 qubits (L=6).
We confirm this hypothesis even for a 24 qubits system (L=8) with close values of N by confronting $r_{252} = 6$, $r_{253} = 110$, $r_{255} = 4$.

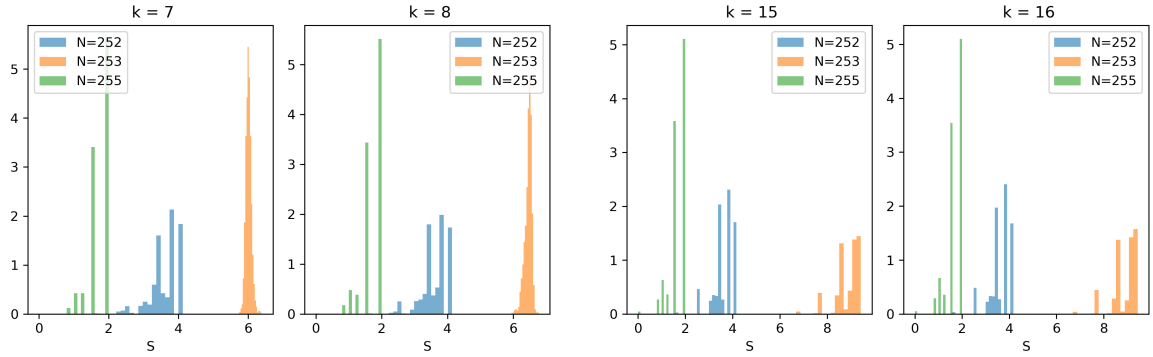


Figure 8.7: L = 8: $r_{252} = 6$,$r_{253} = 110$, $r_{255} = 4$

The distributions generated indeed looks more regular as r increases as we can see in Figure 8.7.
For a 21 qubits system, in Figure 8.8 we confront distributions for the same orders $r_{66} = 10$, $r_{69} = 11$ as in Figure 8.6.
The two sets of distributions in Figure 8.8 have a very different behaviour both between them and with their same ordered counterpart for a $L = 6$ qubits system.
We notice that the distribution for $N = 66$ it's peaked around specific values for $9 \leq k \leq 14$. This means that higher values of the order allow for more delocalized

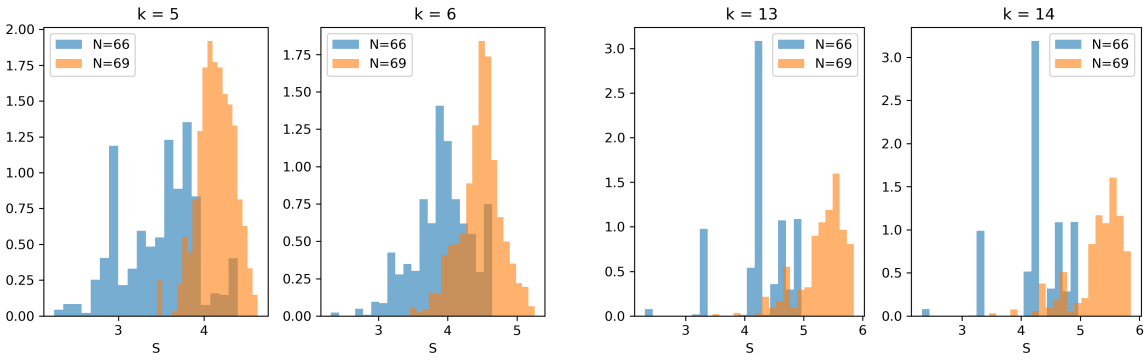


Figure 8.8: L = 7: $r_{66} = 10$,$r_{69} = 11$

distributions, but highly concentrated distributions may still be encountered even for $r \geq 9$.

8.1.4 High order for many qubits system

We now consider the entropy dynamics for a system of 24 qubits and same high ordered pairs $r_{217} = r_{248} = 30$.

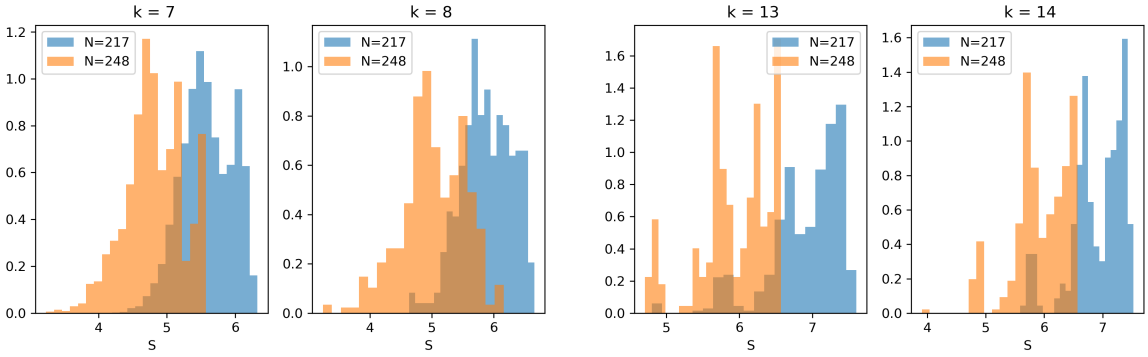


Figure 8.9: L = 8, same order $r = 30$

Even considering systems of 24 qubits and higher values of N, the order is not sufficient to characterize the distributions behaviour: the distributions evolution, behaviour and symmetric properties are very different even for systems with same coprime Y, same order r and same number of qubits $3L$, as we can see in Figure 8.9.

9. Conclusions

In this work we simulated Shor's algorithm on a classical computer and we computed the entanglement entropy for an arbitrary balanced bipartition using parallel and memory efficient construction of the reduced density matrix. We used GPU accelerated methods for computing the spectral decomposition of Hermitian matrices. We evaluated the entropy for every balanced bipartition of the system for up to 18 qubits; for bigger systems we resorted to Montecarlo methods, which we then proceeded to validate. We tested the use of iterative methods for computing the partial spectrum of sparse density matrices in order to approximate the entanglement entropy for high dimensional systems, but our results suggest this solution cannot be considered accurate.

We first introduced the analytically tractable case of entanglement between registers, where the entanglement entropy only depends on the order r of the input numbers Y and N.

Our results suggest that in Shor's algorithm the mean entanglement with respect to all balanced bipartitions of qubits grows across computational steps as a sigmoid function. The sigmoid parameters depend mainly on r, but the maximum value is subject to slight oscillations for different values of N. We also observe that entanglement becomes more multipartite as the order r grows: since r is $O(N)$, we can conclude that, on average, entanglement entropy becomes more multipartite as the input number N increases.

We also modeled the behaviour of entanglement entropy at the end of the modular exponentiation by averaging between all coprimes Y of each N. We observed that, after normalizing with respect to the number of considered qubits, the bipartition dividing the register has an entanglement entropy which is an upper bound for the mean entropy of the balanced bipartitions. Despite this, the entanglement entropy per qubit between registers grows more slowly on average.

The study of entanglement across an algorithm is useful for increasing the accuracy on nowadays NISQ (Noisy Intermediate Scale Quantum) devices: since real life quantum computers have limited fidelity and are not closed systems, the study of entanglement

can help minimizing errors and optimizing physical allocation of qubits in a NISQ device for the specific algorithm.
Usually, qubits on NISQ devices are not interchangeable: the level of noise and the fidelity of gates on different qubits may vary. A quantum device often provides an estimate of external noise, gate fidelity for each qubits and a model of qubits interaction, therefore having a model for the entanglement between qubits in an algorithm allow us to allocate the physical qubits in the most efficient way, i.e. the one that minimizes the error probability.
Since in this work we provided a model and computed the entanglement entropy between qubits, future works may run Shor's algorithm on a NISQ device and measure the entanglement entropy. Then, we could use the experimental data in order to confront our results with the performance of noisy quantum computers when a perturbation is applied.
For example, one could apply a controlled perturbation one qubit at a time and study how the results are affected: according to our data, we expect multipartite entanglement to prevail in circuits with higher order r, therefore we suggest that a single qubit perturbation can affect the majority of the system qubits. We don't expect this to happen for $r \leq 6$.

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A. Entanglement entropy for a partition in the target register

We are actually able to compute analytically the entanglement entropy of a subsystem of the target register: if we consider the target reduced density matrix ρ_k^T in Equation 4.2, we can further trace away some target qubits.
Let us define as E the set of target qubits to trace away, $E \subseteq \{0, .., L-1\}$ with cardinality $|E|$ and let the cardinality of the other subset be $2^L - 2^{|E|} = n$
We start from

$$\rho_k^T = \frac{1}{2^k} \begin{bmatrix} t_0 & 0 & \cdots & & \\ 0 & t_2 & 0 \cdots & 0 & \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 \cdots & t_{2^L-1} & \end{bmatrix} \tag{A.1}$$

We recall that $t_i \neq 0 \iff i \in \{Y^m \bmod N,\, m = 0, 1, ..., r-1\}$ and t_i represents the number of repetitions of i in the modular exponentiation sequence for $m = 0, .., 2^{2L} - 1$.
For sake of simplicity, we change the ordering of the qubits such that qubits in E are last: this still leads us to a diagonal density matrix A_k^T whose diagonal elements we will denote by $a_0, .., a_{2^L-1}$.
We stress that $a_i = t_j$ for some pair of indexes i, j: we have only changed qubits ordering in order to simplify further notation.
The reduced density matrix for the qubits not in E is defined as

$$\rho_{red} = \mathrm{Tr}_E(A_k^T) = \sum_{e=0}^{2^{|E|}-1} \langle e \otimes \mathbb{1}_n |\, A_k^T \,| e \otimes \mathbb{1}_n \rangle \tag{A.2}$$

If we consider $\langle e \otimes \mathbb{1}_n|$, we can see

$$\langle e \otimes \mathbb{1}_n| = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

where the first vector has dimension $2^{|E|}$ and its only nonzero element is a 1 in position $2^{|E|} - e - 1$. Thus, if we use block matrices,

$$\langle e \otimes \mathbb{1}_n| = \begin{bmatrix} 0 & 0 & \cdots & 0 & \mathbb{1}_n & 0 & \cdots & 0 \end{bmatrix}$$

where 0 denotes an all-zero matrix of dimension $n \times n$ and the overall matrix is $n \times 2^L$. We can rewrite

$$\langle e \otimes \mathbb{1}_n| = \sum_{i=0}^{n-1} |i\rangle \langle 2^L - (e+1)n + i| \equiv \sum_{i=0}^{n-1} |i\rangle \langle s_e + i|$$

and similarly

$$|e \otimes \mathbb{1}_n\rangle = \sum_{l=0}^{n-1} |s_e + l\rangle \langle l|$$

Inserting these relationships into a term of the sum in Equation A.2 we get

$$\begin{aligned} &\sum_{i=0}^{n-1} |i\rangle \langle s_e + i| \left(\sum_{k=0}^{2^L-1} a_k |k\rangle \langle k| \right) \sum_{l=0}^{n-1} |s_e + l\rangle \langle l| \quad \implies \delta_{s_e+i,k} \\ &= \sum_{i=0}^{n-1} a_{s_e+i} |i\rangle \langle s_e + i| \sum_{l=0}^{n-1} |s_e + l\rangle \langle l| \quad \implies \delta_{s_e+i,s_e+l} = \delta_{i,l} \\ &= \sum_{i=0}^{n-1} a_{s_e+i} |i\rangle \langle i| \end{aligned}$$

Then, by considering the sum over the basis $\{|e\rangle\}_e$ we get

$$\rho_{red} = \sum_{i=0}^{n-1} \left(\sum_{e=0}^{2^{|E|}-1} a_{s_e+i} \right) |i\rangle \langle i| \tag{A.3}$$

which implies that ρ_{red} is itself a diagonal matrix, where

$$\rho_{red}[i,i] = \sum_{e=0}^{2^{|E|}-1} a_{s_e+i} = \sum_{j=0}^{2^{|E|}-1} a_{j\cdot n} \equiv \hat{a}_i$$

Therefore we have entanglement entropy

$$S(\rho_{red}) = \sum_{i=0}^{n-1} -\hat{a}_i \log_2(\hat{a}_i) \in [0, \min(|E|, k+L-|E|)]$$

It is important to notice that since we traced away first the control register and then E, this entropy represent the entanglement between the qubits in the control register which do not belong in E and all the other qubits, contained in control register or E.
Therefore, since the entanglement entropy is lockable and non additive, we cannot use this relationship in order to properly quantify the entanglement within the control register.